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ANN-integrated modeling of HTL-free Cs₂SnI₆ perovskite solar cells under indoor and outdoor light spectra

Md Zannatul Arif^a, Guobing Zhou^{a,*}, Md. Munirul Hasan^b^a School of Energy, Power and Mechanical Engineering, North China Electric Power University, Beijing 102206, China^b Faculty of Computing, University Malaysia Pahang Al-Sultan Abdullah, Pekan, Pahang 26600, Malaysia

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ABSTRACT

To address the cost and technical challenges in the commercial scalability of perovskite solar cells (PSCs), we propose an HTL-free PSC device with the structure FTO/TiO₂/Cs₂SnI₆/Ni. In this study, we employ affinity engineering methodology to select the most appropriate ETL materials for this device. A range of ETL materials, including SnO₂, CdS, GO, TiO₂, MZO, and WO₃, are evaluated. Among these, TiO₂ shows the best performance and is selected as the optimal ETL material. The HTL-free device is investigated under two light spectra: solar light (outdoor) and LED light (indoor). Key parameters, such as absorber thickness, absorber defect density, electron affinity, and work function of back contact metal, are optimized through simulations. The final optimized device structure—FTO/TiO₂/Cs₂SnI₆/Ni—achieves an impressive efficiency of 27.01 % under solar light and 38.75 % under LED light. Additionally, an Artificial Neural Network (ANN) model is used to predict the power conversion efficiency (PCE). The input data are obtained from SCAPS simulations, and the ANN model is designed with a 4–12–4 architecture. The predictive model is able to estimate the efficiency with a mean squared error (MSE) of 8.127×10^{-4} , a root mean squared error (RMSE) of 0.00944, and a R² value of 0.968 under the solar light spectrum. For the LED light spectrum, the MSE is 3.6585×10^{-4} , the RMSE is 0.00614, and the R² is 0.986. The sensitivity analysis reveals that the absorber thickness is the most significant parameter influencing the device performance under present conditions. The hybrid model demonstrates a strong predictive ability, providing an efficient technique for optimizing the device investigated.

1. Introduction

As global priorities shift toward a greener, more sustainable future, perovskite solar cells (PSCs) are gaining significant attention as a promising power source for alternative energy solutions [1]. The efficiency of PSCs has been rapidly improved from 3.2 % in 2009 to 26.21 % in 2024, positioning them as a leading renewable energy technology with the potential to revolutionize sustainable energy systems [2,3]. Despite these advancements, PSCs still face challenges in achieving high performance and commercial viability [4]. Researchers are exploring various approaches, including numerical modeling and chemical methods, to enhance the efficiency and stability of PSC materials, thus advancing their potential for large-scale applications.

Material optimization is a key to overcoming these efficiency and stability challenges. Recently, lead-free tin-based double perovskite solar cells (DPSCs), particularly Cs₂SnI₆, have shown promise for improving power conversion efficiency (PCE). Cs₂SnI₆ features a

favorable bandgap range of 1.25–1.62 eV, excellent absorption, and high carrier mobility, making it an attractive material for high-efficiency solar cells [5]. Additionally, Cs₂SnI₆ demonstrates excellent air stability due to the presence of Sn⁴⁺, a highly stable oxidation state that resists further chemical degradation [6,7]. This stability, combined with its favorable optoelectronic properties, makes it an exceptional candidate for photovoltaic applications.

Qiu et al. [8,9] were the first to use Cs₂SnI₆ as an absorber material in experiments, achieving PCE of 0.96 % and 0.87 %. Although they employed various materials for the electron transport layer (ETL) and hole transport layer (HTL), including ZnO, TiO₂, and P3HT, the achieved efficiencies were far below the requirements for high-performance photovoltaic devices. Despite ongoing efforts, experimental results with Cs₂SnI₆ have generally remained unsatisfactory [10–16]. To address these efficiency challenges, recent studies have focused on developing Cs₂SnI₆-based PSC device structures. Umedov et al. [17] fabricated an inverted structure with a device configuration of

* Corresponding author.

E-mail address: zhougb@ncepu.edu.cn (G. Zhou).<https://doi.org/10.1016/j.jalcom.2025.181801>

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ITO/CuI/Cs₂SnI₆/PCBM/AZO/Ag, achieving efficiencies of 0.135 % and 0.66 %. Pal et al. [18] investigated an n-i-p planar structure with TiO₂/Cs₂SnI₆/Cu₂O, achieving a power conversion efficiency (PCE) of 17.77 %. Similarly, Chen et al. [19] reported a PCE of 30.39 % using an FTO/IGZO/Cs₂SnI₆/Cu₂BaSnS₄/Au PSC device.

The efficiency and cost-effectiveness of PSC structures are critically influenced by the selection and optimization of the ETL, which plays significant roles in determining the device performance [20]. The ETL materials SnO₂, CdS, GO, TiO₂, MZO, and WO₃ have been extensively applied due to their excellent bandgap, electrical, and optical properties [21–26]. Dong et al. [27] studied a single-layer SnO₂ deposited via the sol-gel method and achieved an efficiency of 7.3 %. Similarly, Tong et al. [28] fabricated a CdS ETL using the chemical vapor deposition method, achieving an efficiency of 9.9 %. Mahmoud et al. [29] experimentally investigated GO-TiO₂ composite nanofiber-based ETLs and reported a remarkable efficiency of 20.06 %. In 2021, Mohammed et al. [30] employed ultraviolet-visible absorption spectroscopy to study WO₃/TiO₂ bilayers, achieving an efficiency of 14.7 %. On the other hand, MZO has been identified as a highly resistive transparent layer in CdTe solar cells, achieving a record efficiency of 20.17 % [31]. Among these ETL materials, TiO₂ holds a prominent position due to its versatility, making it applicable in various solar cell technologies, including energy storage, water treatment through photocatalysis, and PSCs [32–34].

The development of PV structures depends on various factors, including the choice of ETL and HTL materials, absorber materials, and the light spectrum. While significant research has been conducted using the solar spectrum, fewer studies have focused on indoor light spectra. For

instance, Chen et al. [35] used fluorescent lamp spectrum illumination in their ITO/PEDOT: PSS/MAPbI₃-xCl_x/PCBM/Ag PSC structure, achieving a PCE of 27.4 %. On the other hand, Borah et al. [36] investigated three different indoor spectra in their device structure: light emitting diode (LED) light, compact fluorescent light (CFL), and incandescent (INC) light, achieving PCE values of 40.08 %, 39.22 %, and 29.15 %, respectively.

Optimizing device performance under different light spectra presents significant challenges. In recent years, various machine learning models, such as artificial neural networks (ANNs), support vector machines (SVMs), random forests (RFs), and XGBoost (XGB), have been widely applied in photovoltaic research [37–39]. Particularly, ANNs have proven to be a powerful tool for optimization issues. ANNs can efficiently analyze large datasets, recognize patterns, and predict photovoltaic performance across various lighting conditions, thus accelerating the design process and enhancing the ability to tailor device configurations for both outdoor and indoor environments [36]. Unlike traditional experimental methods, ANNs guide predictions by identifying promising configurations, reducing reliance on time-consuming experiments [40].

ANN models are capable of predicting key parameters like power conversion efficiency (PCE), open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), and fill factor (FF). For example, Li et al. [41] used the ANN model to predict the bandgap of perovskite materials, complementing both experimental and simulation work. By capturing complex patterns in data, these models streamline device modeling and simulation processes. ANNs, in particular, offer higher prediction accuracy, flexibility, and efficiency compared with traditional methods, making them invaluable in enhancing solar cell performance and advancing material design strategies.

Recent studies on perovskite solar cells (PSCs) have extensively focused on optimizing hole transport layers (HTLs) to enhance both power conversion efficiency and operational stability. For example, Azhakanantham et al. [42] investigated suitable HTL materials for FASnI₃-based PSCs, reporting improved charge transport and overall device performance. Similarly, Karuppaiah et al. [43] proposed molybdenum disulfide (MoS₂) as a sustainable alternative to conventional

Spiro-OMeTAD, achieving notable improvements in efficiency and environmental compatibility. Although these HTL-based strategies show significant promise, they often involve complex fabrication processes and elevated material costs, which may limit their scalability and commercial viability. In contrast, this study proposes a cost-effective HTL-free PSC design, using Cs₂SnI₆ as the absorber and TiO₂ as the ETL. Eliminating the HTL simplifies device configuration and fabrication while maintaining robust performance under both outdoor (AM1.5 solar spectrum) and indoor (LED) lighting conditions.

Device performance is first analyzed via SCAPS-1D simulations, followed by ANN modeling. To validate the proposed structure, the study follows a systematic approach: (1) Validation of the HTL-free model; (2) Comparison of the HTL-free PSC (FTO/ETL/Absorber/Au) with the conventional design (FTO/ETL/Absorber/HTL/Au); (3) Investigation of various ETL materials (SnO₂, CdS, GO, TiO₂, MZO, WO₃) using affinity engineering to identify the optimal ETL; (4) Evaluation of critical parameters such as absorber thickness, defect density, electron affinity, and metal work function; (5) Prediction of device efficiency using the ANN model based on SCAPS-1D simulations. This integrated approach highlights the potential of HTL-free PSCs for both indoor and outdoor photovoltaic applications. To the best of our knowledge, few prior studies have combined SCAPS-1D simulations with ANN modeling to optimize HTL-free Cs₂SnI₆-based PSCs under both solar and indoor lighting conditions.

2. Theoretical framework and modeling

2.1. SCAPS-1D simulation methodology

SCAPS-1D has been utilized in this study to investigate the performance of solar cell devices by analyzing their electrical and optical parameters [44–46]. This widely used simulation tool is particularly effective for exploring configurations like n-i-p and p-i-n planar structures, enabling a detailed examination of device layers and their impact on overall efficiency. By leveraging SCAPS-1D, this study aims to optimize and enhance the proposed solar cell structure. SCAPS-1D operates by solving three fundamental semiconductor equations:

2.1.1. Poisson's equation

Poisson's equation describes the spatial distribution of the electric potential (ϕ) within the solar cell. It relates the divergence of the electric field to the local charge density

$$\frac{d^2\phi}{dx^2} = -\frac{\rho}{\epsilon} \quad (1)$$

where, ϕ and x represent the electrostatic potential (V) and position (nm), respectively. ϵ is permittivity of the material (F/m). ρ is the net charge density (C/m³), which is defined as

$$\rho = q(p - n + N_D^+ - N_A^-) \quad (2)$$

Here, q is elementary charge (C), p and n present concentrations of holes and electrons (cm⁻³). The ionized dopants, N_D^+ and N_A^- , represent the concentrations of donor and acceptor dopants, respectively, (cm⁻³).

2.1.2. Continuity equation for electrons

The continuity equation for electrons ensures the conservation of electron charge within the solar cell. It represents the rate of change in electron concentration, considering the flow of electron current as well as the processes of generation and recombination. The general form is

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + (G_n - R_n) \quad (3)$$

where, $\frac{\partial n}{\partial t}$: Rate of change of electron concentration over time;

J_n : Electron current density (A/m²);

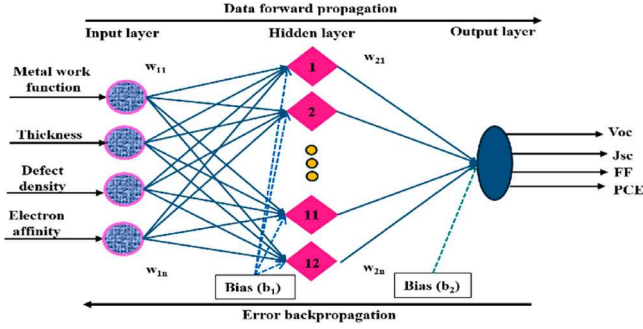


Fig. 1. Network configuration of the applied model, structured as 4–12–4, with four input neurons, a single hidden layer containing twelve neurons, and four output neurons.

G_n : Electron generation rate ($\text{cm}^{-3}\text{s}^{-1}$);
 R_n : Electron recombination rate ($\text{cm}^{-3}\text{s}^{-1}$).

2.1.3. Continuity equation for holes

The continuity equation for holes ensures the conservation of hole charge within the solar cell, which is similar to the electron equation. It represents the rate of change in hole concentration, considering the flow of hole current and the processes of generation and recombination. The general form is

$$\frac{\partial p}{\partial t} = \frac{1}{q} \frac{\partial J_p}{\partial x} + (G_p - R_p) \quad (4)$$

where, $\frac{\partial p}{\partial t}$: Rate of change of hole concentration over time;

J_p : Hole current density (A/m^2);

G_p : Hole generation rate ($\text{cm}^{-3}\text{s}^{-1}$);

R_p : Hole recombination rate ($\text{cm}^{-3}\text{s}^{-1}$).

The simple optical investigation on the proposed PV structure can be conducted using the SCAPS software. By default, the software is set to use the standard solar spectrum ($1000 \text{ W}/\text{m}^2$) with air mass 1.5 G, which allows for a comprehensive simulation of solar cell performance under typical sunlight conditions. On the other hand, when considering spectra from LED sources, the software considers the total power density (W/m^2) emitted by the LEDs the number of photons absorbed by a material is determined by the incident wavelength λ . This relationship governs the generation of electron-hole pairs and is mathematically expressed as

$$G(x) = \int_{\lambda_{\min}}^{\lambda_{\max}} a(\lambda, x) N_{\text{phot}}(\lambda, x) d\lambda \quad (5)$$

where, $G(x)$ is the generation rate of electron-hole pairs ($\text{cm}^{-3}\text{s}^{-1}$);
 x represents the depth (nm);

λ is the wavelength (nm);

$a(\lambda, x)$ is the absorption coefficient (cm^{-1});

$N_{\text{phot}}(\lambda, x)$ is the photon flux ($\text{photons}\cdot\text{cm}^{-1}\cdot\text{nm}^{-1}$);

$\lambda_{\max}$ and $\lambda_{\min}$ define the spectral range of incident photons (nm).

This formulation enables the simulation software to analyze the impact of different light sources on the performance of PV structures by considering variations in spectral distribution and photon energy absorption.

2.2. ANN framework for predictive analysis

Recently, ANN models have been widely utilized as predictive tools, particularly in feed-forward configurations for forecasting and analysis. ANNs are trained using a method called backpropagation, which adjusts weights and biases to minimize errors in predictions. An ANN model typically consists of one input layer, one or more hidden layers, and one output layer. Each layer contains interconnected nodes (neurons) that process and transmit information. All input and output values are normalized between -1 and 1 to ensure uniformity and improve the performance of the ANN model. This normalization is generally achieved using a standard equation:

$$y = \frac{(y_{\max} - y_{\min})(x - x_{\min})}{x_{\max} - x_{\min}} + y_{\min} \quad (6)$$

where, x is the Input value in the original range $[x_{\min}, x_{\max}]$. y is the normalized value in the target range $[y_{\min}, y_{\max}]$. $y_{\min}$ and $y_{\max}$ are the

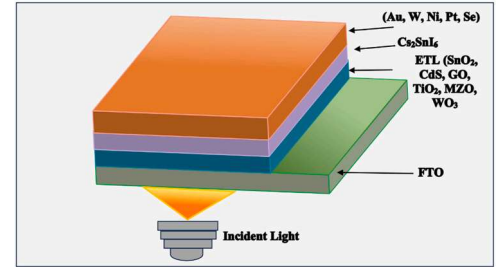


Fig. 3. Proposed HTL-free PSC device.

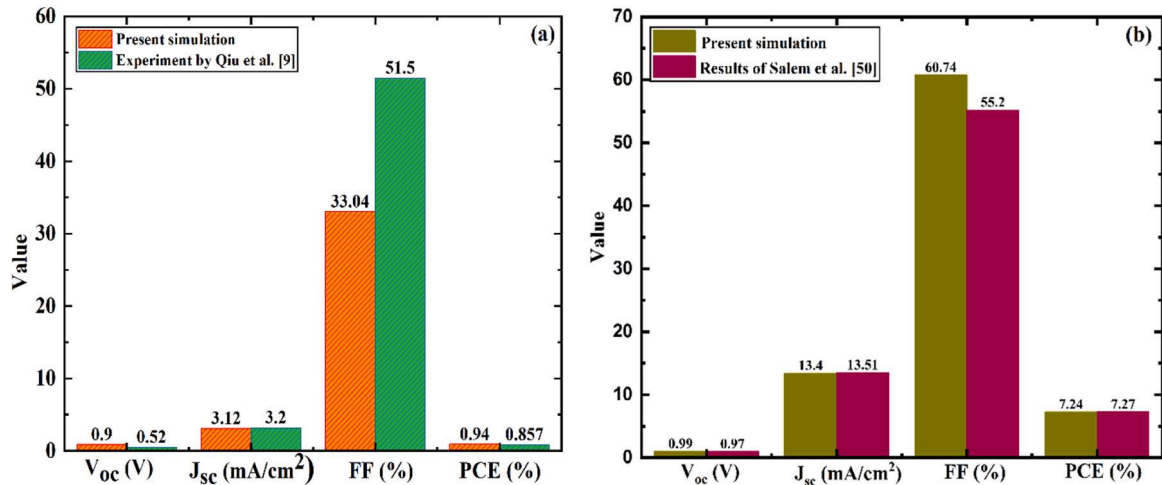


Fig. 2. Validation of the model: (a) n-i-p structure comparison, (b) detailed results of the HTL-free model.

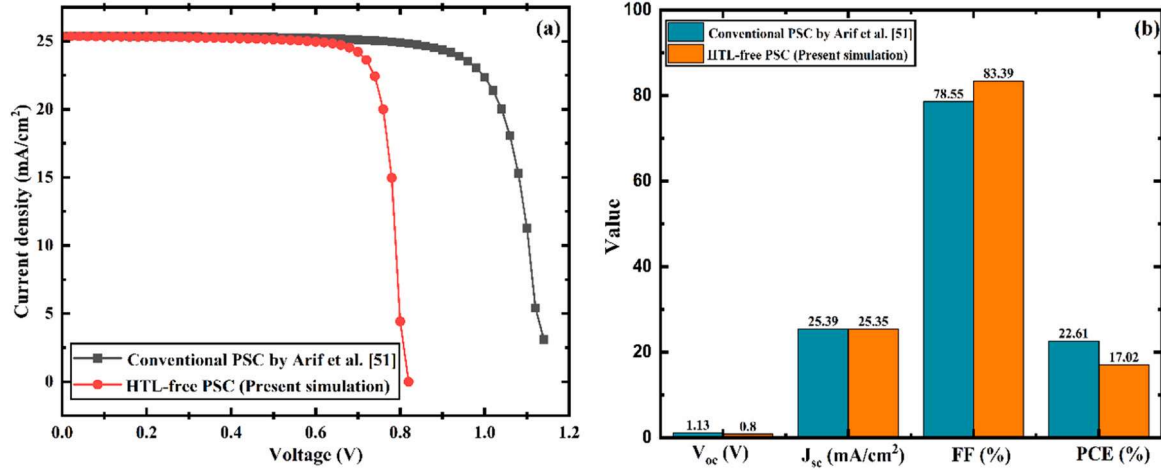


Fig. 4. Comparison of conventional and HTL-free devices: (a) J - V characteristics, (b) detailed results.

Table 1

Main parameters of the PSC device.

Parameters	FTO [52,54]	(SnO ₂) [49,53]	TiO ₂ [56]	GO [57]	CdS [55]	MZO [58]	WO ₃ [59]	Cs ₂ SnI ₆ [8,9]
Thickness T (nm)	200	50–100	50–100	50–100	50–100	50–100	50–100	100–1500
Band gap E_g (eV)	3.50	3.4	3.20	3.0	2.4	3.3	2.6	1.48
Electron affinity χ (eV)	4	4	4.20	3.86	4.18	4	3.8	4.01
Dielectric permittivity ϵ_r (eV)	9	9	9	9.0	10	66	4.8	10.0
Electron mobility μ_e (cm ² V ⁻¹ s ⁻¹)	20	20	100	1.0×10^2	25	0.5	30	53
Hole mobility μ_h (cm ² V ⁻¹ s ⁻¹)	10	10	25	3.0×10	100	0.5	30	3.1×10^{-2}
C_B density of states N_c (cm ⁻³)	2.2×10^{18}	2.2×10^{17}	2.0×10^{17}	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}	2.2×10^{21}	2.2×10^{18}
V_B density of states N_v (cm ⁻³)	1.8×10^{19}	2.2×10^{16}	1.0×10^{17}	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}	2.2×10^{21}	1.8×10^{19}
Shallow donor density N_D (cm ⁻³)	2×10^{19}	1×10^{18}	1.0×10^{17}	1.0×10^{15}	1×10^{18}	1×10^{18}	1×10^{18}	0
Shallow acceptor density N_A (cm ⁻³)	0	0	0	0	0	0	0	1.0×10^{14}
Defect density N_t (cm ⁻³)	1×10^{15}	1.0×10^{14}	1×10^{14}	1.0×10^{14}	1×10^{15}	1.0×10^{14}	1.0×10^{14}	1.0×10^{14}
Defect type	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral	Neutral
Capture cross section electron (cm ²)	10^{-15}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}
Capture cross section holes (cm ²)	10^{-15}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}	2×10^{-14}
Energetic distribution	Gauß	Gauß	Gauß	Gauß	Gauß	Gauß	Gauß	Gauß
Reference for defect energy level E_t	Above	Above	Above	Above	Above	Above	Above	Above
	E_v	E_v	E_v	E_v	E_v	E_v	E_v	E_v
Energy level with respect to reference (eV)	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Characteristic energy (eV)	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1

desired minimum and maximum normalized values. $x_{\min}$ and $x_{\max}$ are the minimum and maximum values in the original data range.

An ANN neuron can be mathematically represented as a basic model in which the input value (p_i) is multiplied by the i^{th} weighting factor of the k^{th} neuron. This value is summed up by the bias (b_k), and σ is the activation function. Then, an ANN neuron can be expressed mathematically, as in Eq. 7.

$$y_k = \sigma \left(\sum_{i=1}^n w_{ki} p_i + b_k \right) \quad (7)$$

In this research, to evaluate the effectiveness of the proposed ANN model, performance metrics coefficient of regression (R^2), mean square error (MSE), and root mean square error (RMSE) is used. Eq. (8) ~ (10) present the mathematical formula of the R^2 , MSE and RMSE.

$$MSE = \frac{1}{m} \sum_{j=1}^m (v_{SCAPS} - v_{ANN})^2 \quad (8)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{j=1}^m (v_{SCAPS} - v_{ANN})^2} \quad (9)$$

$$R = \frac{\sum [(v_{SCAPS} - \lambda_{SCAPS}) * (v_{ANN} - \lambda_{ANN})]}{\lambda_{SCAPS} * \lambda_{ANN}} \quad (10)$$

where m is the number of data samples, v_{SCAPS} represents the simulation result from SCAPS, and v_{ANN} denotes the predicted outcome from the ANN model. RMSE provides a measure of the error magnitude in the same units as the data. λ_{SCAPS} and λ_{ANN} represent the mean values of v_{SCAPS} and v_{ANN} , respectively.

The learning process of the ANN model is divided into two main stages: forward propagation and backpropagation. During forward propagation, the weights and biases are initially assigned with random values. These parameters are used to calculate the output of the hidden layers and eventually the output layer, based on the input data. The dataset is obtained from SCAPS-1D simulations, with 70 % allocated for training, 15 % for testing, and 15 % for validation. This split was chosen to ensure that the model had sufficient data to learn underlying patterns (training set), while also allowing for performance monitoring during training (validation set) and unbiased evaluation of unseen data (testing set). The data was shuffled before splitting to ensure a uniform distribution of input features and target outputs across all subsets, thereby preventing any bias related to the order of the data. This rigorous data preparation strategy aims to develop a well-generalized ANN model capable of accurately predicting outcomes for new, unseen input conditions. The model's input parameters include thickness, electron affinity, and metal work function properties, while the outputs are V_{oc} , J_{sc} , FF, and PCE under both light spectra. All input and output values are normalized within the range of -1 to 1 to maintain uniformity and

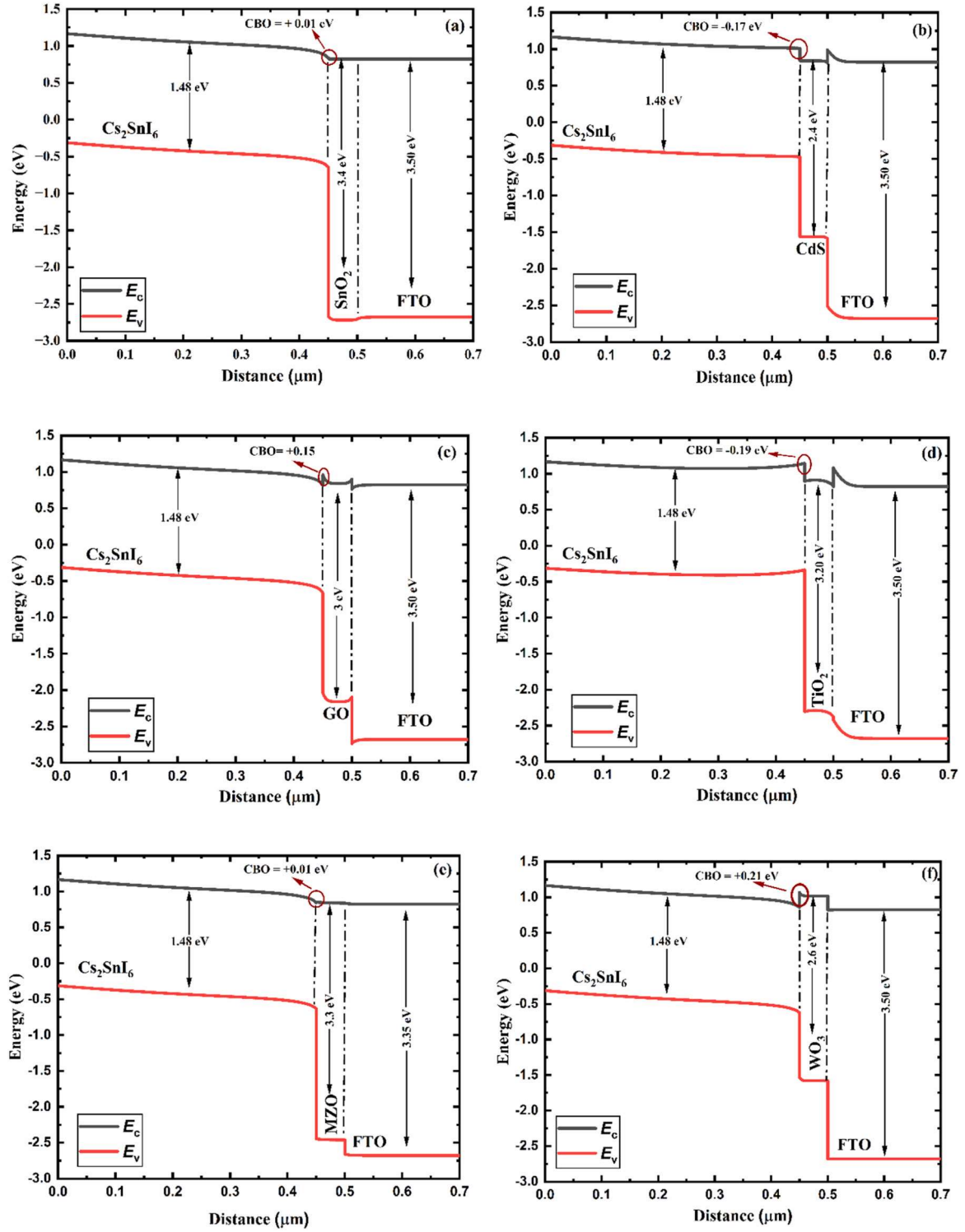


Fig. 5. Energy band diagrams of HTL-free PSC configurations (a–f) for band alignment analysis, where E_c represents the conduction band offset and E_v represents the valence band offset.

enhance the performance of the ANN model.

The network structure is determined through a trial-and-error approach, with the final architecture consisting of a single hidden layer with 12 neurons, as depicted in Fig. 1. The model is trained using the Levenberg-Marquardt (LM) algorithm, with a learning rate of 0.01. The activation functions used are tansig in the hidden layer and purelin in the output layer. The corresponding activation functions are as follows:

$$\text{tansig} : f(x) = \frac{1}{1 + e^{-x}} - 1 \quad (11)$$

$$\text{purelin} : f(x) = 1 \quad (12)$$

The tansig activation function, which is based on the sigmoid function, is used in the hidden layer, mapping the input values to a range of $[-1, 1]$. Meanwhile, the purelin activation function, which is a linear

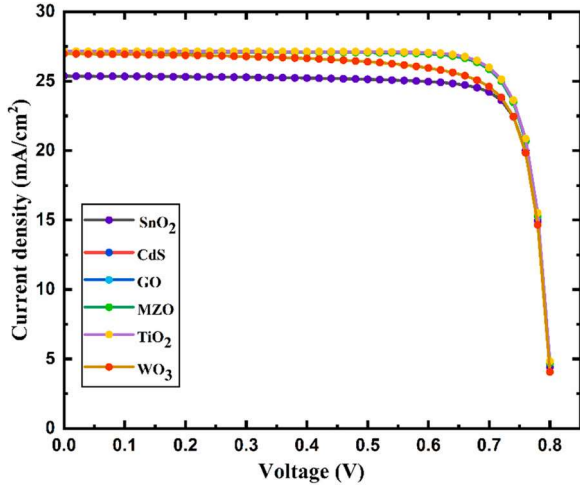
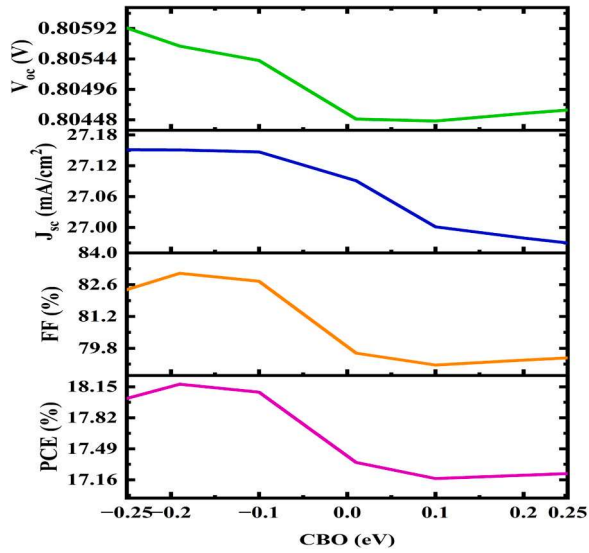
Fig. 6. J - V characteristics of different ETL materials.

Fig. 7. Effect of CBO on the efficiency of PSCs.

function, is applied in the output layer.

2.3. HTL-free model validation

To validate the accuracy of the proposed HTL-free model, we compare its performance with both experimental and simulated results reported in the literature. Specifically, we investigate two structures: an experimental n-i-p model (FTO/ZnO/Cs₂SnI₆/P3HT/Au) and present simulation of an HTL-free model (FTO/CdZnS/Sb₂S₃/Au). We use literature data from references [9,47–52], including studies by Qiu et al. [9] and Salem et al. [50]. Fig. 2(a) shows a comparison between the present simulation results and the experimental data reported by Qiu et al. [9]. The simulated PCE is 0.94 %, which closely matches the experimental value of 0.87 %, resulting in a relative deviation of approximately 8.05 %. Fig. 2(b) presents a broader performance comparison with the results reported by Salem et al. [50], where the relative deviation in PCE is approximately 0.41 %, further supporting the accuracy of the present model. These findings validate the accuracy of the proposed model, confirming its reliability for this study based on the literature.

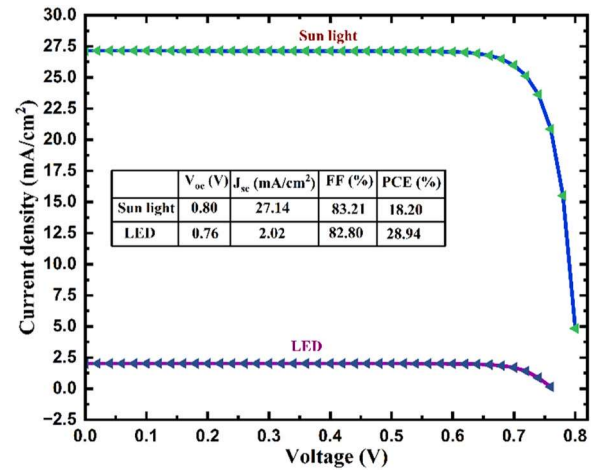
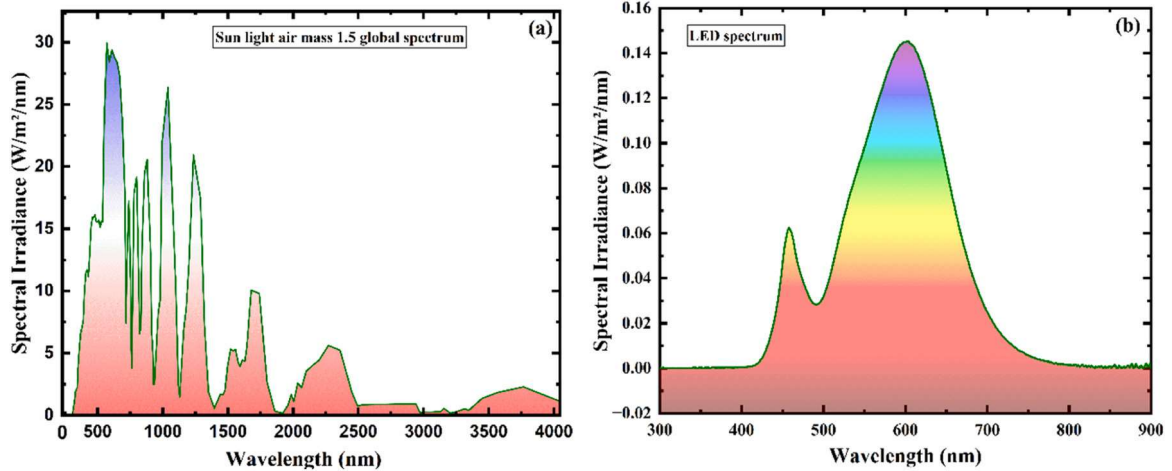
Fig. 9. J - V characteristics based on different light sources.

Fig. 8. Different sources of light spectrum: (a) Sun light; (b) LED light.

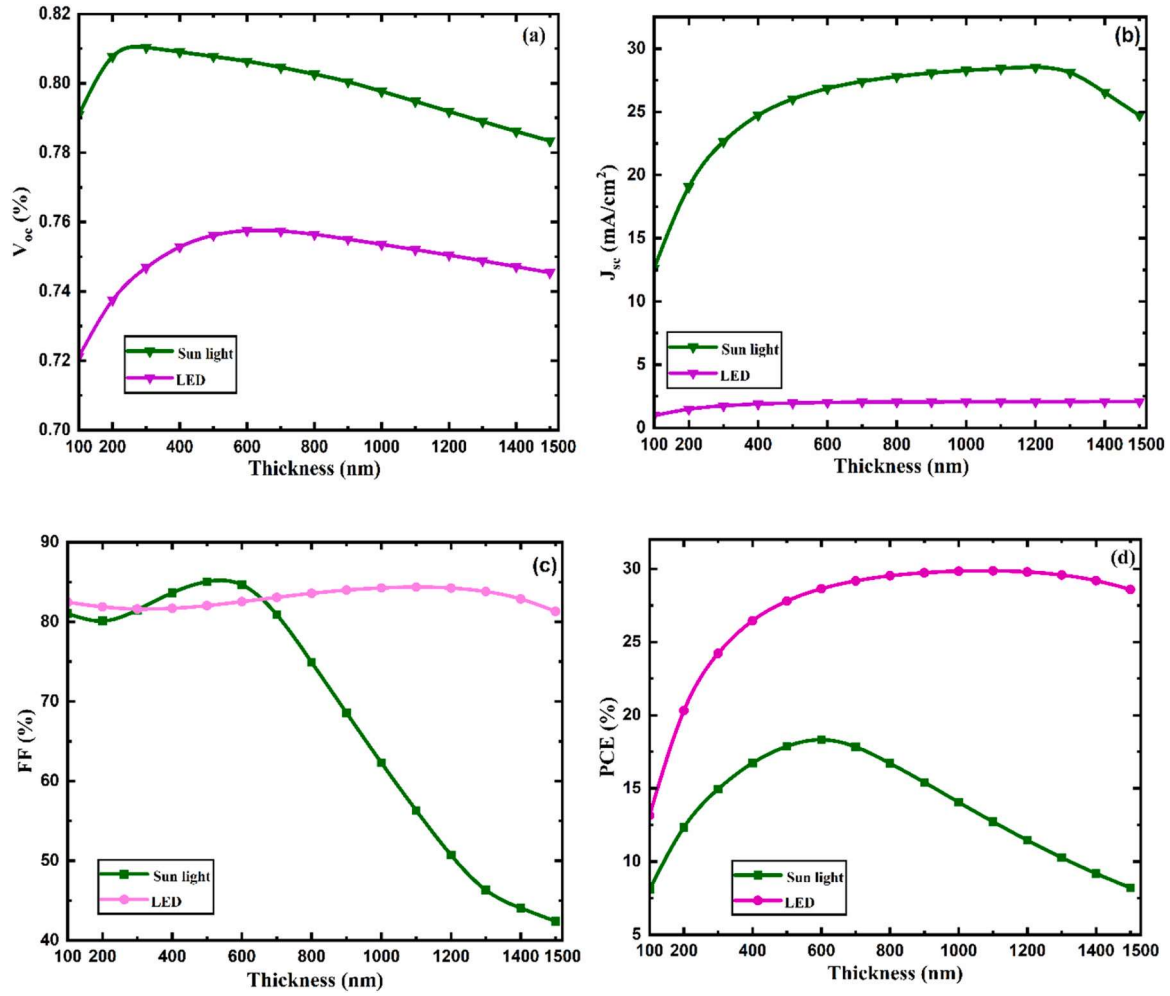


Fig. 10. Variation of performance metrics with the absorber thickness under sunlight and LED light sources.

3. Results and discussion

3.1. Proposed PSC device structure and comparative analysis of HTL-free vs. conventional models

Earlier, we proposed an FTO/SnO₂/Cs₂SnI₆/MoO₃/Au device structure and achieved a PCE of 22.60 % [51]. To address the cost challenges, we propose a modified HTL-free PSC device structure: FTO/ETL/Cs₂SnI₆/Au (Fig. 3), where FTO serves as the glass substrate, Cs₂SnI₆ as the absorber and Au as the back contact. To enhance the efficiency of the HTL-free device, various ETL materials, including SnO₂, CdS, GO, TiO₂, MZO, and WO₃, are evaluated to identify the most suitable one for this PV device. For the back contact, other materials like W, Ni, Pt, and Se are also considered in addition to Au.

Fig. 4(a) illustrates the J - V characteristics comparing the conventional PSC device with the present HTL-free device. Data from the literature [8,9,52–55] are utilized to design both devices. The J_{sc} values for the two devices are nearly identical, with the conventional device achieving 25.35 mA/cm² and the HTL-free device achieving 25.39 mA/cm². However, the voltage of the HTL-free device is lower, at 0.8 V, compared to 1.13 V for the conventional one. Fig. 4(b) shows the detailed performance results of both devices. It is seen that the FF difference is significant, with the HTL-free device achieving 83.39 % and the conventional one achieving 78.55 %. Consequently, the PCE of the HTL-free device (17.02 %) is lower than that of the conventional device (22.60 %). The lower efficiency of the HTL-free device is attributed to the absence of the hole transport layer, which affects the charge

collection efficiency. To overcome this limitation, various strategies, such as optimizing the ETL material and utilizing the electron affinity technique, are explored in subsequent sections.

3.2. Impact of ETL optimization on HTL-free PSC performance based on conduction band offset (CBO)

The ETL and HTL enable the directed flow of electrons and holes, respectively, ensuring efficient transfer to the cathode and anode while preventing undesired electron flow to the metal contacts. The interface between the ETL and absorber, or HTL and absorber, is crucial for minimizing recombination and improving performance. To reduce costs, we have eliminated the HTL layer, focusing on a device with only the ETL layer. Our primary focus is evaluating the impact of electron affinity on the performance of HTL-free devices. The ideal ETL is selected based on the conduction band offset (CBO), which significantly affects device performance. The following equation is used to calculate the CBO:

$$CBO = \chi_{\text{absorber}} - \chi_{\text{ETL}} \quad (14)$$

where χ is defined as the electron affinity for Cs₂SnI₆ and ETL materials, which can be obtained from Table 1. We have processed the band diagram for the FTO/ETL/absorber/Au device using data from Table 1. Fig. 5 represents the band alignment of the six different ETL materials, where the valence band (E_v) represents the energy levels where electrons are typically located in their ground state, and the conduction band (E_c) represents the energy levels where electrons can freely move and contribute to electrical conduction.

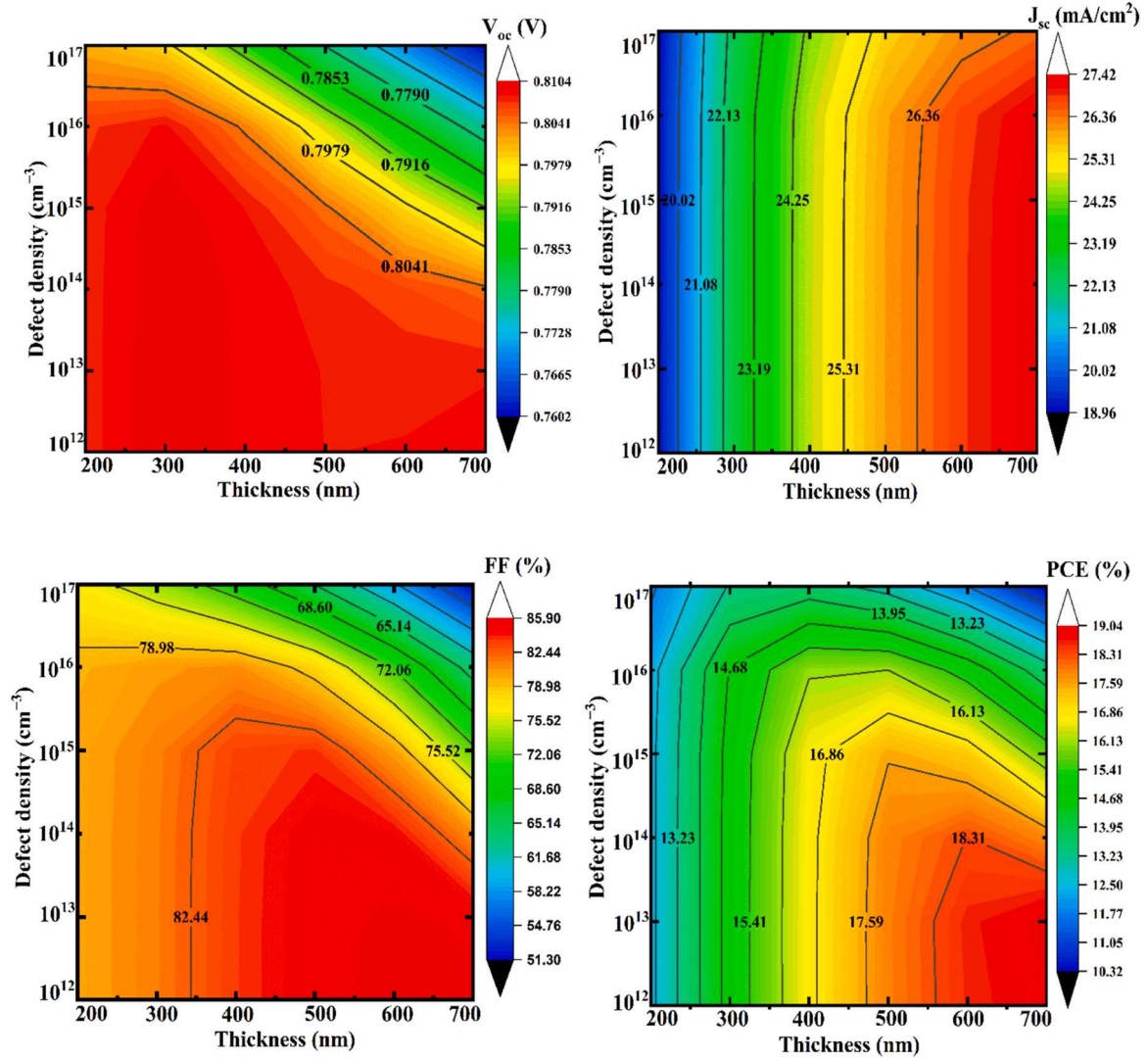


Fig. 11. Variation of performance metrics with the absorber defect density under sunlight.

Typically, the CBO occurs in three types: cliff-like, spike-like, and flat-like [50]. Fig. 5(a-d) presents the different scenarios of CBO structures. Fig. 5(a) illustrates the Cs₂SnI₆/SnO₂ band alignment structure, with a CBO of approximately +0.01 eV, indicating minimal band bending, resulting in a nearly flat-like structure. This structure suggests that the electron barrier is nearly zero, allowing for efficient charge transport [60]. Fig. 5(b) depicts the Cs₂SnI₆/CdS alignment with a cliff-like structure and a CBO of -0.17 eV, indicating a negative CBO. This negative CBO supports effective electron transport with overall good performance. Fig. 5(c) shows the Cs₂SnI₆/GO alignment, characterized by a spike-like structure and a CBO of +0.15 eV. This alignment might also contribute to increased interface recombination due to carrier trapping at the interface. Fig. 5(d) represents the Cs₂SnI₆/TiO₂ alignment, which also exhibits a cliff-like structure but with a CBO of -0.19 eV. Fig. 5(e) highlights the Cs₂SnI₆/MZO alignment with a nearly flat-like band structure and a CBO of +0.01 eV, resulting in an almost zero electron charge barrier.

Finally, Fig. 5(f) presents the Cs₂SnI₆/WO₃ alignment with a spike-like structure and a CBO of +0.21 eV. Among the materials considered, CdS and TiO₂ are both promising candidates due to their negative CBOs and cliff-like band structures. These materials facilitate efficient electron transport while minimizing recombination. The negative CBO (cliff-like structure) promotes smooth electron flow from the perovskite layer to the ETL [61]. At the same time, the band alignment helps to

reduce electron trapping at the interface. This improves the overall performance of the device by ensuring efficient electron extraction and minimizing loss mechanisms [62].

Fig. 6 illustrates the J - V characteristics of the PSC with different ETL materials under the AM 1.5 sunlight spectrum. The J - V characteristics reveal slight variations in performance. The current density values range between 25.36 mA/cm² and 27.14 mA/cm², while the maximum voltage observed remains consistent at approximately 0.82 V. These differences can be attributed to the variations in charge transport efficiency and the energy band alignment of the ETL materials [63]. To ensure optimal energy conversion, it is crucial to analyze the I - V characteristics and fine-tune the operating conditions to align with the maximum power point (MPP).

Fig. 7 shows the variations in PV cell performances based on the CBO. It is seen that V_{oc} decreases as the CBO increases. The highest V_{oc} of 0.80593 V is observed at a CBO of -0.25 eV. However, the ETLs, including TiO₂, CdS, MZO, SnO₂, GO, and WO₃, exhibit V_{oc} values ranging from 0.80564 V to 0.80458 V, with CBO values ranging from -0.19 eV to 0.21 eV. For J_{sc} , TiO₂ to GO show similar values ranging from 27.09 mA/cm² to 27.16 mA/cm², while WO₃ has the lowest value at 26.97 mA/cm² at a CBO of 0.21 eV. This trend reflects the impact of CBO on device performance. The FF varies, with TiO₂ achieving the highest value of 83.10 % at a CBO of -0.19 eV. The FFs of other materials like CdS, SnO₂, MZO, GO, and WO₃ range from 79.27. % to

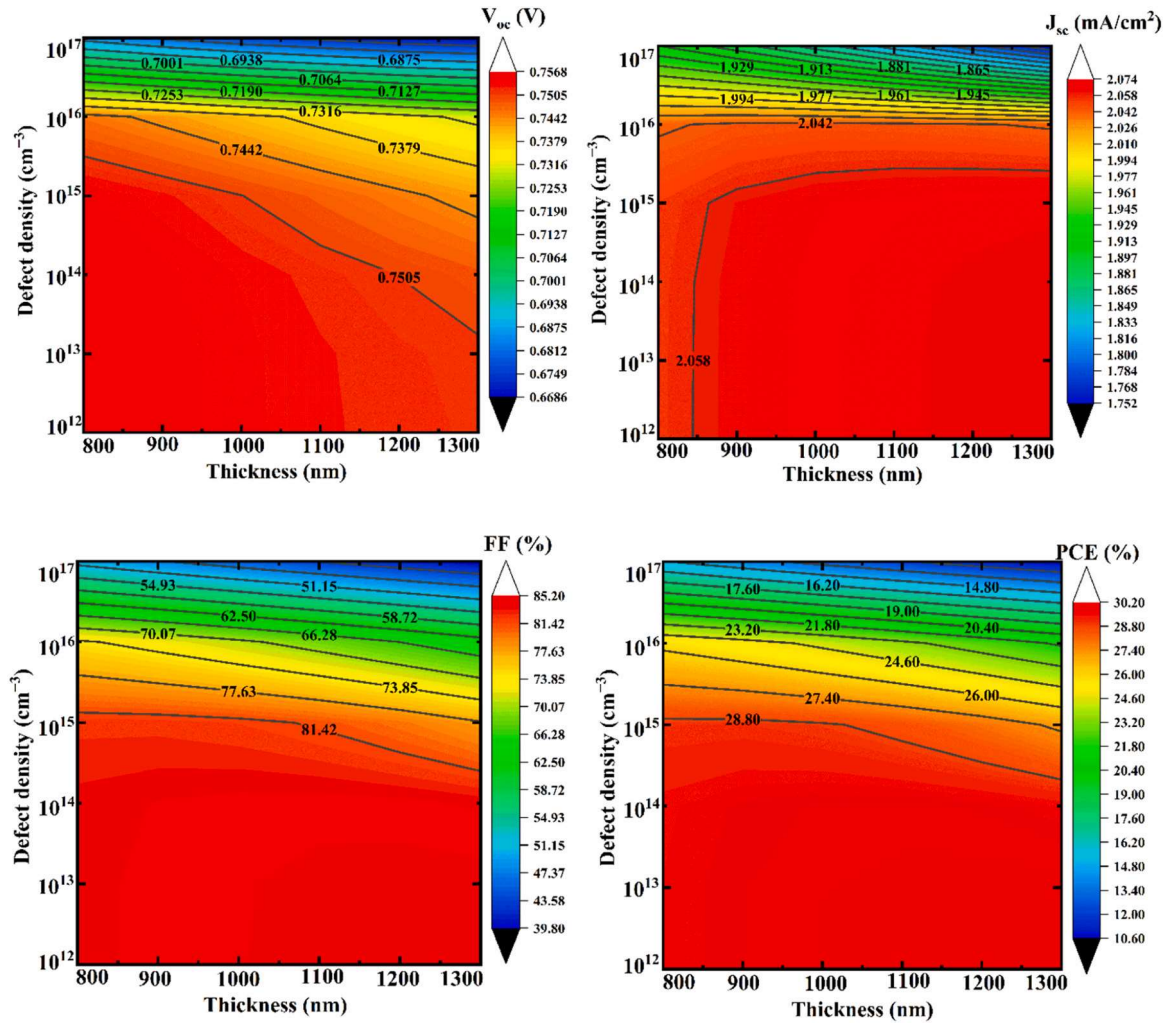


Fig. 12. Variation of performance metrics with the absorber defect density under LED.

82.74 %, with CBO values being from -0.17 eV to 0.21 eV. The highest PCE is observed for TiO_2 at 18.17 % with a CBO of -0.19 eV. Other materials show PCE values as follows: MZO at 17.34 %, GO at 17.17 %, CdS at 18.09 %, WO_3 at 17.20 %, and SnO_2 at 17.20 %. Based on the PCE, TiO_2 is the optimal material, ensuring efficient electron transport, and is selected for the final simulations.

3.3. Impact of the absorber on PSC performance under various light spectra

Photovoltaic (PV) structures are typically evaluated under the standard solar spectrum (AM 1.5 G) to simulate outdoor conditions. In this study, both solar and LED light sources are used to assess performance in outdoor and indoor environments. Fig. 8(a) shows the Solar AM1.5 G spectrum used from the built-in dataset of the SCAPS-1D software. Fig. 8 (b) illustrates the light spectra obtained from previous literature sources [64,65]. The LED setup included red, blue, and white light chips to simulate indoor lighting collected under controlled conditions. The LED spectrum used is directly adopted from these literature sources, ensuring a realistic representation of indoor lighting conditions. The solar spectrum covers wavelengths from 0 nm to 4000 nm with spectral irradiance up to $30 \text{ W/m}^2/\text{nm}$, while the LED spectrum spans 300 nm to 900 nm with irradiance ranging from -0.02 – $0.16 \text{ W/m}^2/\text{nm}$. This dual-spectrum approach provides insights into the PV structure's versatility for outdoor and indoor applications, making it suitable for hybrid energy solutions.

Fig. 9 illustrates the respective J - V characteristics of the PSC structure under the solar and LED light sources. The current density ranges from 2.02 mA/cm^2 under LED illumination to 27.14 mA/cm^2 under solar light. The corresponding voltages are 0.76 V and 0.80 V , respectively. The FF values for LED and solar light are 82.80 % and 83.21 %, while the PCE are observed to be 28.94 % and 18.20 %, respectively. Interestingly, the PCE under indoor LED illumination is higher than that under outdoor solar light. However, the current density is significantly higher under solar light and lower under LED illumination. This variation can be attributed to the differences in spectral intensity and photon flux between the two light sources [66]. Solar light, with its broader spectrum and higher irradiance, facilitates greater current generation. In contrast, the LED spectrum, being narrower and more focused, allows for efficient photon absorption under indoor conditions but results in lower current density due to limited photon availability. Additionally, the higher PCE under LED light can be attributed to improved spectral alignment with the absorber bandgap and suppressed recombination losses due to lower carrier density, which enhances V_{oc} and FF [67]. This behavior is also consistent with previous simulation studies on perovskite solar cells under indoor lighting conditions [36,67,68].

Since the thickness of the absorber may play a critical role in determining the PSC efficiency, the HTL-free device performances are evaluated as a function of the absorber thickness (ranging from 100 nm to 1500 nm) under both LED and solar spectra. Fig. 10 (a) shows the V_{oc} ranges from 0.72 V to 0.75 V under LED illumination and 0.79 V to 0.81 V under sunlight, with the highest values observed at 600 nm for

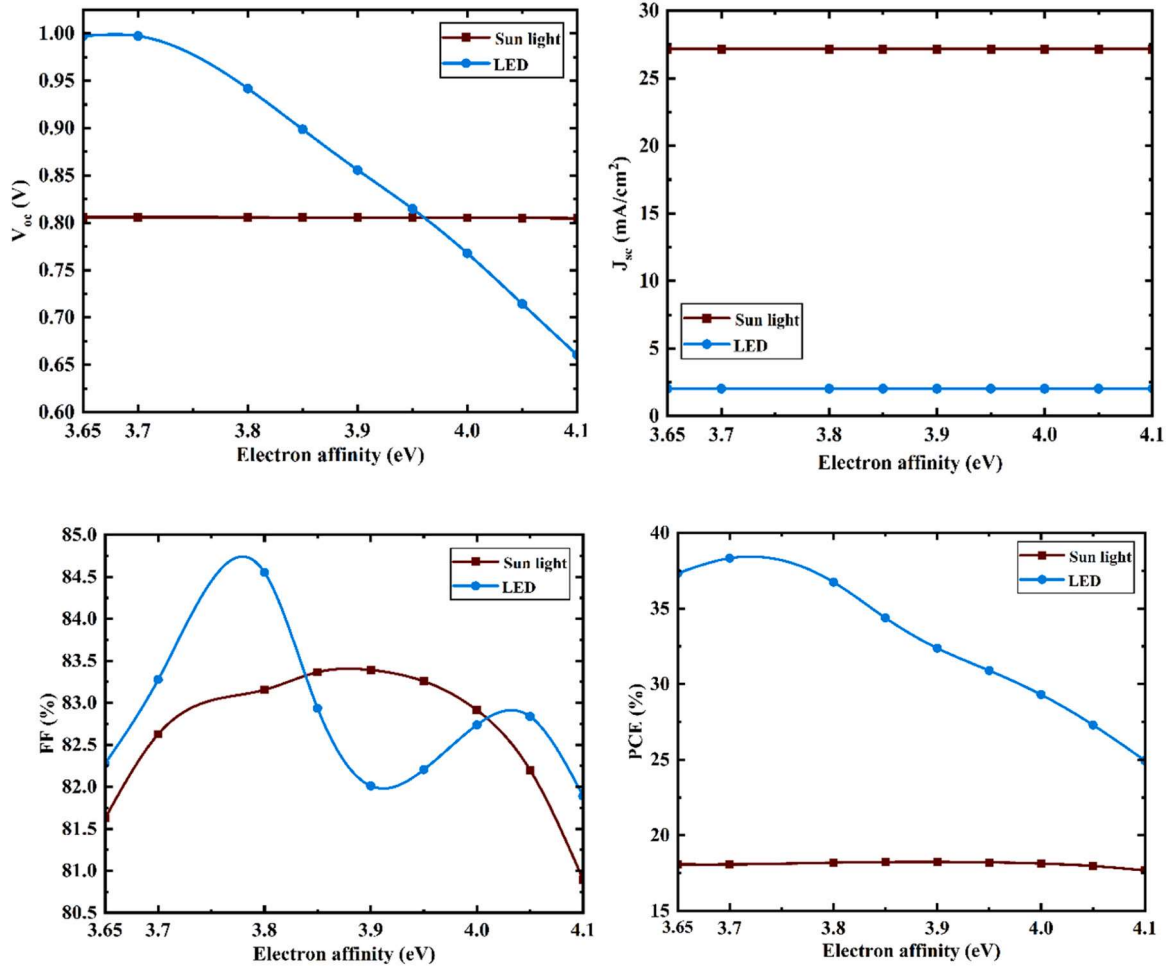


Fig. 13. Variation of performance metrics with the electron affinity under sunlight and LED light sources.

LED light and 300 nm for sunlight, respectively. This variation in V_{oc} is attributed to differences in light absorption and charge carrier dynamics influenced by the spectral properties of the two light sources. Similarly, as shown in Fig. 10 (b), the J_{sc} trends differ significantly. The LED spectrum gives a nearly constant J_{sc} of 2.07 mA/cm² across the thickness range, while the sunlight spectrum initially increases, reaching a maximum of 28.52 mA/cm² at 1200 nm, before decreasing. Fig. 10 (c) demonstrates that the FF under the LED spectrum ranges from 81.31 % to 84.36 % across all thicknesses. In contrast, under sunlight spectrum, the FF peaks at 85.11 % at 600 nm and gradually decreases with increasing thickness. Finally, as shown in Fig. 10 (d), the PCE under the LED spectrum increases with thickness, reaching a maximum of 29.85 % at 1100 nm. For the solar spectrum, the maximum PCE is observed at 18.32 % at 600 nm. The efficiency trends differ due to the spectral characteristics of indoor (LED) and outdoor (solar) light sources. These differences arise from spectral characteristics: under LED light, the optimal thickness balances absorption and charge extraction, whereas under solar illumination, excessive thickness leads to higher recombination losses. It is important to note that while thicker absorber layers enhance photon absorption under broad-spectrum solar light (AM1.5 G), they may not be optimal under LED light. Due to the narrower spectrum and lower intensity of LED light, thinner absorbers may be more effective for reducing recombination losses and improving charge collection efficiency. The 600 nm for solar spectrum and 1100 nm for LED spectrum is used for final device optimization.

3.4. Impact of the absorber defect density on PSC performance under various light spectra

In this section, we investigate the impact of absorber defect density using a contour graph. The defect density varies from 10^{12} cm⁻³ to 10^{17} cm⁻³, while the absorber thickness ranges from 200 nm to 700 nm. A lower defect density effectively reduces recombination and energy losses, thereby enhancing charge transport, while increasing absorber thickness improves the current density. Striking a balance between these parameters is essential for optimizing device efficiency. Fig. 11 presents the contour mapping of defect density under standard sunlight spectra. The V_{oc} ranges from 0.8104 V to 0.7602 V, with minimal variation. The highest V_{oc} is recorded at a defect density of 10^{12} cm⁻³ and a thickness of 300 nm. Similarly, the highest J_{sc} of 27.41 mA/cm² is achieved at a defect density of 10^{12} cm⁻³ and a thickness of 700 nm. The FF exhibited significant variation, ranging from 51.30 % to 85.90 %. The highest FF of 85.90 % is observed at a defect density of 10^{12} cm⁻³ and a thickness of 700 nm. The highest PCE of 19.04 % is recorded at a defect density of 10^{13} cm⁻³ and an absorber thickness of 600 nm. Overall, this investigation highlights the crucial role of defect density in PSC performance, demonstrating that a lower defect density and a thicker absorber layer improve efficiency. Based on the PCE results, the optimal defect density for the final device is determined to be 10^{13} cm⁻³ for sunlight.

Fig. 12 presents the impact of defect density under an LED spectrum. The defect density varies from 10^{12} cm⁻³ to 10^{17} cm⁻³, while the absorber thickness ranges from 800 nm to 1300 nm. The V_{oc} using the LED spectrum ranges from 0.66 V to 0.75 V, with the highest V_{oc} of 0.75 V recorded at a defect density of 10^{12} cm⁻³ and a thickness of

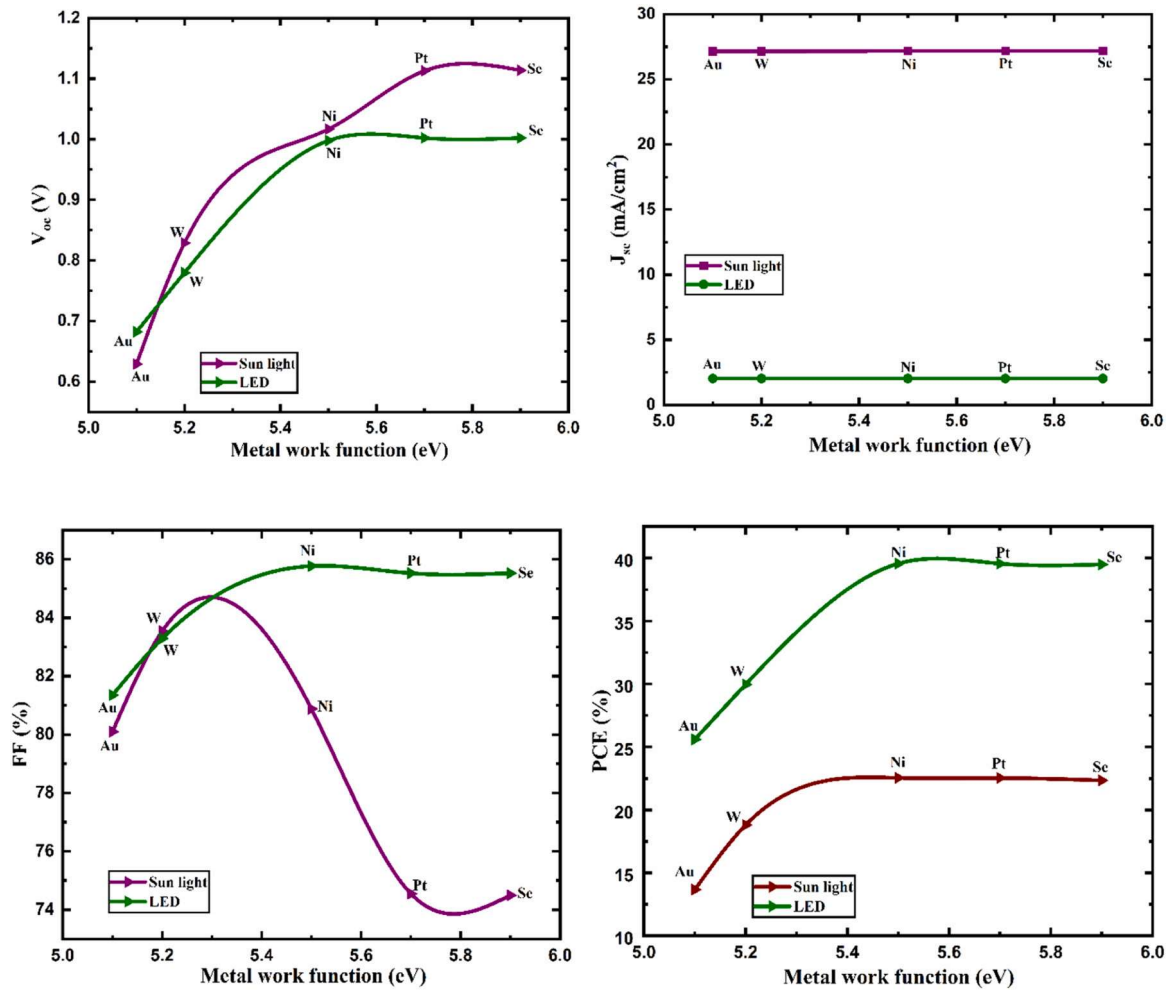


Fig. 14. Variation of performance metrics with the back contact metal under sunlight and LED light sources.

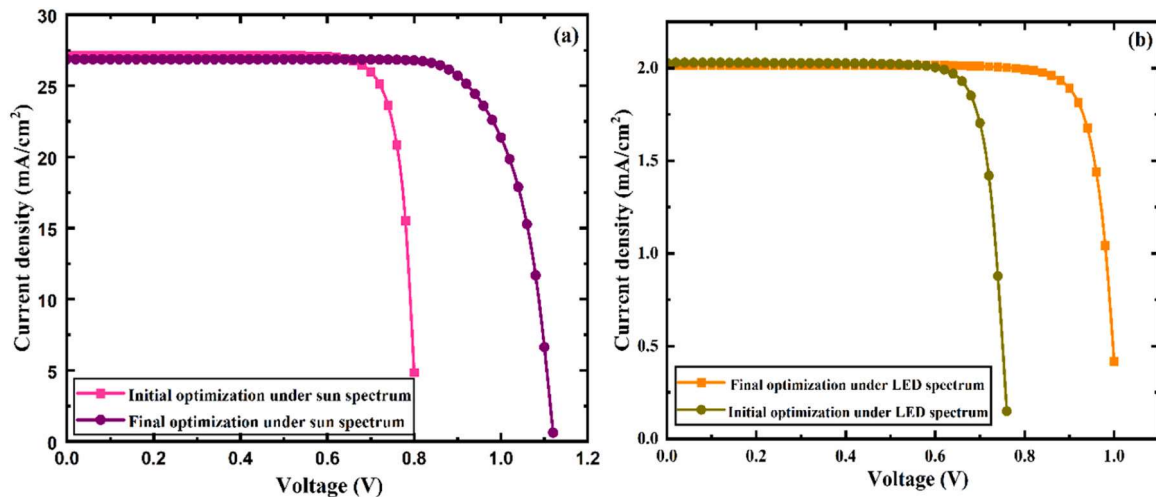


Fig. 15. J - V characteristics for initial and final optimization: (a) Sun spectrum, (b) LED spectrum.

800 nm. In contrast to sunlight spectra, the J_{sc} under the LED spectrum is significantly lower due to the spectral characteristics of LED lighting, which provides a narrower wavelength range and lower photon energy. The highest J_{sc} value of $2.07 \text{ mA}/\text{cm}^2$ is achieved at a defect density of 10^{12} cm^{-3} and a thickness of 1300 nm. However, the FF exhibited the highest value of 85.07 % observed at 1300 nm and a defect density of

10^{13} cm^{-3} . The PCE also demonstrated promising results, with the highest PCE of 30.07 % recorded at a defect density of 10^{13} cm^{-3} and a thickness of 1100 nm. Interestingly, the PSC showed better overall performance under LED illumination than under sunlight. This is likely due to the reduced thermal stress and lower recombination losses, as LED lighting minimizes heat-induced degradation and provides a stable

Table 2
Initial and optimized performance of HTL-free PSC under various light spectra.

Simulation conditions for HTL-free Device	V _{oc} (V)	J _{sc} (mA/cm ²)	FF (%)	PCE (%)
(FTO/TiO ₂ /Cs ₂ SnI ₆ /Au) Initial structure for LED spectrum	0.7576	2.029187	82.80	28.94
(FTO/TiO ₂ /Cs ₂ SnI ₆ /Au) Initial structure for solar spectrum	0.8055	27.149630	83.21	18.20
(FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni) Final optimization structure for LED spectrum	1.0097	2.01	83.72	38.75
(FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni) Final optimization structure for solar spectrum	1.11	26.86	82.26	27.01

operating condition. Based on these observations, a defect density of 10¹³ cm⁻³ is selected for the final simulation of the LED spectrum.

3.5. Impact of the electron affinity on PSC performance under various light spectra

Electron affinity plays a critical role in modifying the energy levels essential for the efficiency of PSCs, directly impacting charge transfer and recombination losses. In this study, we focus on an HTL-free device structure, where interactions occur directly at the ETL/absorber interface. This design alters the energy band alignment, which may potentially enhance the device's efficiency by optimizing charge transport at the interface. The absence of HTL makes this alignment even more critical, as it governs electron injection, charge carrier extraction, and recombination processes. Therefore, electron affinity and the alignment of the ETL and absorber layer are key considerations for understanding charge dynamics in HTL-free PSCs.

Fig. 13 illustrates the electron affinity performance matrices. Under the sunlight spectrum, the V_{oc} is 0.805 V, while for the LED spectrum, it increases to 0.997 V at an electron affinity of 3.7 eV. The V_{oc} for the solar spectrum remains relatively stable across the electron affinity range from 3.7 eV to 4.1 eV. In contrast, the V_{oc} for the LED spectrum shows a rapid decrease as the electron affinity increases from 3.7 eV to 4.1 eV. Regarding the J_{sc}, there are minimal changes across the electron affinity range for both spectra. The highest J_{sc} observed under the sunlight spectrum is 27.149 mA/cm², while the LED spectrum shows a value of 2.03 mA/cm². Notably, the maximum J_{sc} for the sunlight spectrum occurs at 4 eV, while for the LED spectrum, it is observed at 3.65 eV. Molecules with high electron affinity enhance hole extraction, improving both the FF and J_{sc} in PSCs. The FF and PCE show distinct trends. For the sunlight spectrum, the FF reaches its peak at 83.39 % at 3.9 eV, and it changes rapidly across the 3.7 eV to 4.1 eV range. The LED spectrum, on the other hand, exhibits oscillating FF values, with the highest value of 84.55 % achieved at 3.8 eV. The PCE for the sun light peaks at 18.23 % at 3.9 eV, while for the LED spectrum, it reaches 38.32 % at 3.7 eV. Based on these results, an electron affinity of 3.9 eV is selected for the sunlight spectrum, and 3.7 eV is chosen for the LED spectrum in the final simulation. For the cell operating under both conditions, the electron affinity can be chosen as a compromise, either as an average or based on the dominant spectrum [66]. The optimization of

Table 3
Comparison of HTL-free Cs₂SnI₆ device with other Cs₂SnI₆-based structures.

PSC structure of Cs ₂ SnI ₆	Investigation type	V _{oc} (V)	J _{sc} (mA/cm ²)	FF (%)	PCE (%)	Ref.
FTO/TiO ₂ /Cs ₂ SnI ₆ /P3HT/Ag	Experimental	0.256	7.41	24	0.47	[5]
Pt/Electrolyte/Dye/Cs ₂ SnI ₆ /TiO ₂ /FTO	Experimental	0.67	16.3	61.8	6.8	[17]
ITO/ B-c-CsSnI ₃ /Cs ₂ SnI ₆ /Au	Experimental	0.63	6.1	28	1.1	[13]
FTO/TiO ₂ /Cs ₂ SnI ₆ /Cu ₂ O/Au	Simulation	0.74	30	80.5	17.77	[18]
Glass/FTO/GO/Cs ₂ SnI ₆ /Cu ₂ O/Au	Simulation	0.837	34.6	81.64	23.64	[7]
FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni (LED)	Simulation	1.0097	2.01	83.72	38.75	Present work
FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni (sun light)	Simulation	1.11	26.86	82.26	27.01	Present work

electron affinity plays a crucial role under both light conditions. Under solar illumination, a wider range of electron affinity values can generally be tolerated due to the higher photon generation. However, under LED light, precise tuning of the electron affinity is critical for achieving proper band alignment, which is essential for efficient charge extraction due to the lower intensity and narrow spectrum of the light source [68].

3.6. Impact of the work function of back contact metal on PSC performance under various light spectra

To accurately evaluate the performance of a PSC device based on the work function of the back contact metal, it is essential to consider its influence on charge extraction and resistance losses. The back contact determines series resistance, directly affecting the output current and voltage. For long-term stability and efficiency, the back contact should minimize chemical interaction with the perovskite layer to prevent degradation. An optimal back contact metal reduces recombination losses and enhances charge extraction, improving overall device performance. Additionally, it facilitates electron collection from the absorbing interface, further enhancing efficiency [69]. In this study, various back contact metals are investigated under both LED and sunlight spectra. Fig. 14 presents a detailed performance analysis of the back contact materials with corresponding work functions, including

Table 4
Comparison of the Cs₂SnI₆ HTL-free device with other HTL-free PSC structures.

PSC structure	V _{oc} (V)	J _{sc} (mA/cm ²)	FF (%)	PCE (%)	Ref.
ITO/WS ₂ /Sr ₃ Ti ₂ S ₇ /Au	0.32	48.49	72.48	11.19	[70]
ITO/ZnMgo/MASnI ₃ /Ag	0.918	34.48	83.79	26.55	[71]
FTO/ZnOS/CsGel ₃ /W	1.29	23.68	86.75	26.7	[72]
FTO/TiO ₂ /CH ₃ NH ₃ SnI ₃ /Ni	0.98	31.93	84.34	26.33	[73]
FTO/ZnO/BiFeO ₃ /Au	1.03	15.27	75.59	11.92	[74]
FTO/CdS/MASnBr ₃ /Au	0.99	32.31	85.56	27.37	[75]
FTO/CH ₃ NH ₃ SnI ₃ /C60/Au	0.8585	32.45	73.72	20.54	[76]
FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni	1.0097	2.01	83.72	38.75	Present work
FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni	1.11	26.86	82.26	27.01	Present work

Table 5
Performance summary of the ANN model using sunlight spectrum.

Performance metrics	Training	Testing	Validation	Overall
R ²	0.964	0.973	0.959	0.968
MSE	6.3845 × 10 ⁻⁴	2.7341 × 10 ⁻³	7.4839 × 10 ⁻⁴	8.9127 × 10 ⁻⁴
RMSE	0.02526	0.01653	0.00865	0.00944

Table 6
Weights and biases of the ANN model for using sunlight spectrum.

Input layer weights				Input layer biases	Hidden layer weights	Output Bias
Input 1	Input 2	Input 3	Input 4			
−1.9935	−3.6223	3.1009	−1.650807	2.1527	−0.0103	−0.4279
−0.6325	2.477	4.4963	1.514409	−1.9612	−0.4906	
−4.2054	−2.4105	1.7623	1.738344	2.3094	0.0121	
2.3844	1.0193	−3.8991	−2.034376	−3.1339	−1.8247	
4.3274	−1.2112	−2.7265	0.224777	−2.0241	−0.4384	
−3.1378	2.3568	−1.6779	−1.084259	2.842	−3.0231	
0.1545	−4.1811	−4.0129	0.601195	0.0883	0.0417	
−4.0454	2.442	−3.5778	−0.947446	−1.4262	−3.2740	
3.0885	3.0606	0.836	−0.190852	0.6293	−0.4796	
2.8119	0.9641	2.6254	−1.529947	1.0509	−2.0260	
−4.3392	3.1163	−1.4533	−2.904263	0.82	−0.3203	
5.0809	−1.599	−1.2703	−2.655437	0.4246	−0.0676	

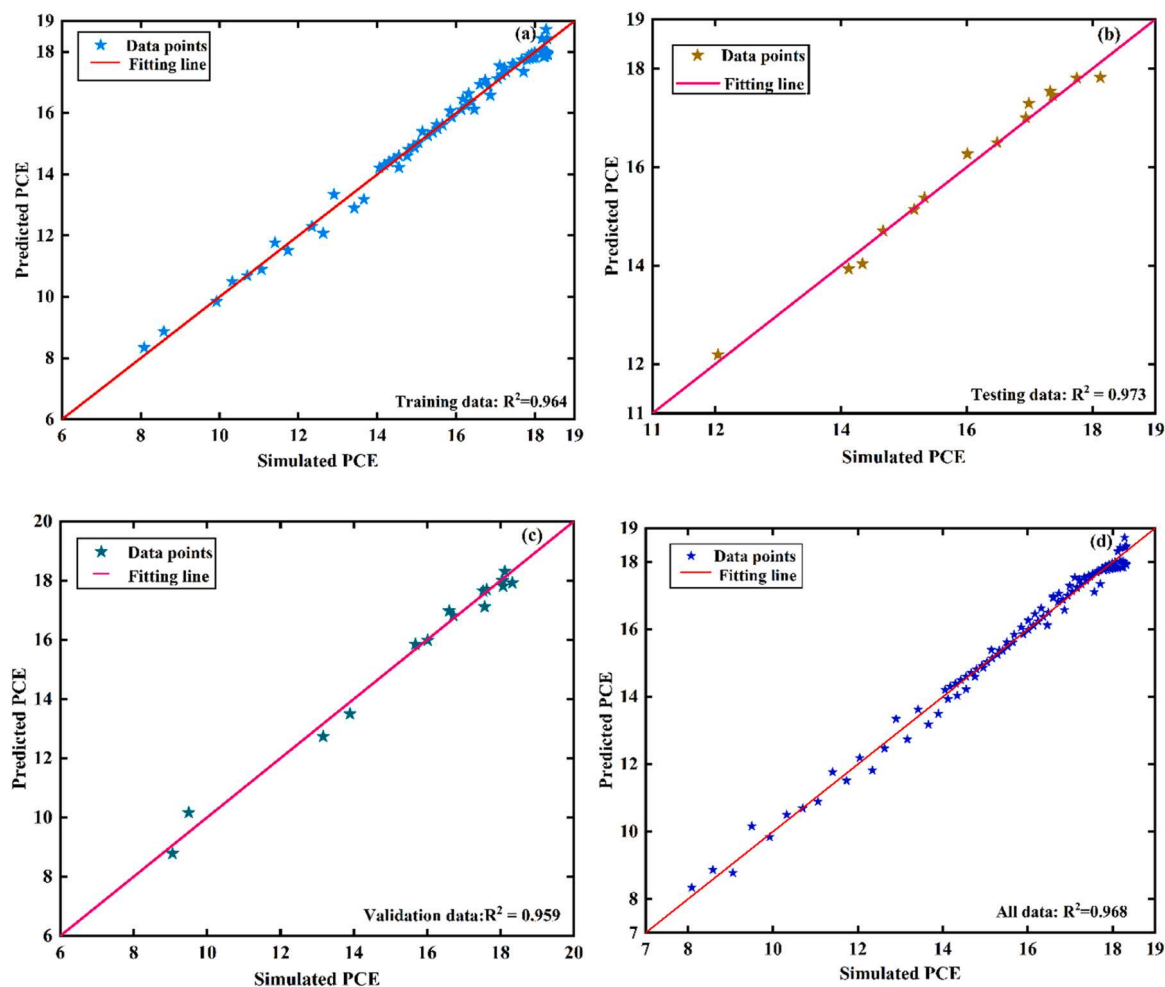


Fig. 16. Regression plots between simulated PCE and ANN-predicted PCE under the sun light spectrum: (a) training, (b) testing, (c) validation, and (d) all data.

gold (Au, 5.1 eV), tungsten (W, 5.2 eV), nickel (Ni, 5.5 eV), platinum (Pt, 5.7 eV), and selenium (Se, 9.9 eV). The V_{oc} exhibits a consistent trend across both light spectra. V_{oc} increases with the work function of the back contact, reaching maximum values for Pt of approximately 1 V under LED light and 1.1 V under sunlight. In contrast, the J_{sc} remains unchanged across the range of metal work functions, from Au to Se, which is approximately 2 mA/cm² under LED light and 27.12 mA/cm² under sunlight. This indicates that current density is not significantly influenced by the metal work function. The FF demonstrates distinct behavior depending on the light spectrum. Under solar light, FF ranges from 74 % to 84.8 %, with the highest value observed for W. Under LED

light, FF varies between 81.2 % and 85.2 %, with Ni achieving the highest value. The PCE is also spectrum-dependent. The maximum PCE is observed for Ni, reaching 40 % under LED light and 22.79 % under sunlight. Both light sources identify Ni as the optimal metal for back contact, which is then selected for the final simulation. Ni has a high work function, which supports stable device operation by forming an ohmic contact in the PV structure. Due to this high work function, Ni facilitates efficient charge extraction and reduces energy barriers at the back contact. However, Ni also exhibits increased resistance over extended periods, which can impact the overall performance of the device.

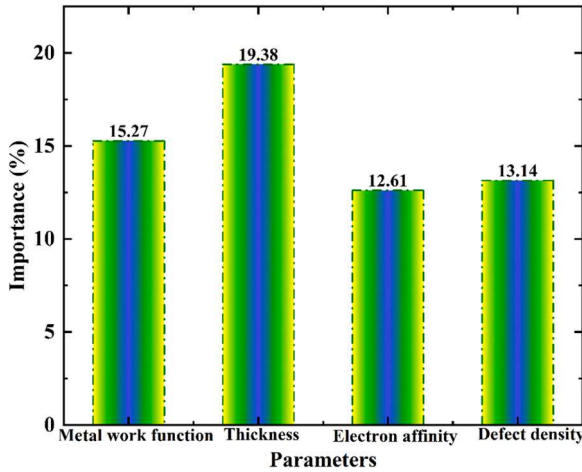


Fig. 17. Parameters importance in the ANN model for sunlight spectrum.

Table 7

Performance summary of the ANN model using LED light spectrum.

Performance metrics	Training	Testing	Validation	Overall
R ²	0.985	0.992	0.978	0.986
MSE	8.6215×10^{-4}	4.6520×10^{-4}	6.5734×10^{-4}	3.6585×10^{-4}
RMSE	0.00928	0.00682	0.00810	0.00614

3.7. Optimized PSC results under LED and sun light spectra

The investigations focus on absorber thickness (T), absorber defect density (N_t), electron affinity (χ), and back contact metal (ϕ_m) to determine the final parameters for the optimized HTL-free PSC device configuration, which is FTO/TiO₂/Cs₂SnI₆/Ni. These investigations, conducted under both LED and sunlight spectra, showcase the device performances for both indoor and outdoor photovoltaic applications. The final parameters above for the HTL-free PSC under the LED spectrum are $T = 1100$ nm, $N_t = 10^{13}$ cm⁻³, $\chi = 3.7$ eV, and $\phi_m = 5.5$ (Ni). Under the sun light spectrum, the final parameters are $T = 600$ nm, $N_t = 10^{13}$ cm⁻³, $\chi = 3.9$ eV, and $\phi_m = 5.5$. Fig. 15 (a) shows the J - V characteristics under the sunlight spectrum, with the initial optimization (FTO/TiO₂/Cs₂SnI₆/Au) yielding a current density of 27.14 mA/cm² at 0.80 V, and the final optimization (FTO/TiO₂/Cs₂SnI₆/Ni) showing 26.86 mA/cm² at 1.11 V. Fig. 15 (b) presents the J - V characteristics under the LED spectrum, where the initial current density of (FTO/TiO₂/Cs₂SnI₆/Au) is 2.02 mA/cm² at 0.75 V, and the final optimization (FTO/TiO₂/Cs₂SnI₆/Ni) gives 2.01 mA/cm² at 1.00 V. Table 2 presents the

detailed optimization results for both initial and final configurations under both spectra. The final optimization yields a PCE of 27.01 % for the solar spectrum and 38.75 % for the LED light spectrum, with both showing improvements over the initial configuration.

Table 3 presents a comparison of the HTL-free Cs₂SnI₆ device with other Cs₂SnI₆-based structures under standard sunlight illumination, including experimental results, simulation results, and findings from our present investigation. The comparison demonstrates that our device achieves the best performance. Similarly, Table 4 compares the HTL-free Cs₂SnI₆ device structure with other HTL-free PSC structures using different materials. Despite these structural differences, our results show significant improvements, making this design promising for future investigations.

3.8. Performance evaluation of the prediction model using ANN

The generalization ability of a trained ANN is crucial for evaluating its predictive performance. Since there is no universally accepted method for selecting the optimal number of hidden neurons, various network configurations are considered and trained [77,78]. The selection of the 4–12–4 ANN architecture was guided by the need to balance model complexity with predictive accuracy. To validate its suitability, several alternative architectures were evaluated using the trial-and-error method, including deeper and wider networks. Although some deeper configurations showed marginal improvements in training performance, they frequently led to overfitting and increased computational cost, with no notable enhancement in validation accuracy. In contrast, the 4–12–4 architecture consistently demonstrated superior generalization, stable training behavior, and computational efficiency. These outcomes were supported by k-fold cross-validation and key performance indicators such as the R² and MSE. Consequently, the 4–12–4 structure was identified as the most reliable and effective model for this application. The final ANN model uses the configuration with 12 neurons for the hidden layer, yields the best performance, and is selected as the final ANN model to predict the efficiency and essential parameters, including thickness, electron affinity, and metal work function, under both LED and solar light spectra for an HTL-free PSC device. Table 5 presents the performance of the ANN model when trained using the sunlight spectrum data. The R-squared value is very close to 1, indicating that the output results closely align with the target values. Additionally, the very low MSE and RMSE values reflect the high accuracy of the ANN model in predicting the efficiency of the HTL-free PSC device. Table 6 also presents the weights, biases, and parameters of the trained ANN model, which can be used for PCE prediction.

A sensitivity analysis was performed to assess the robustness of the ANN model in predicting PCE by adding a small random noise to the input parameters. In this case, Gaussian noise with a mean of zero and a standard deviation of 1–3 % of each input's range was added to simulate measurement uncertainties. The ANN model's performance was then re-

Table 8

Weights and biases of the ANN model for using LED light spectrum.

Input layer weights				Input layer Bias	Hidden layer weights	Output layer biases
Input 1	Input 2	Input 3	Input 4			
-3.3938	-2.14	-3.8042	-0.0572	5.6142	-3.138198	-0.36149
-3.2619	2.1315	-1.9127	2.8801	-5.0423	-2.665985	
-3.9837	2.0547	-0.0524	2.2093	4.0788	-1.603189	
4.0743	2.0138	-2.5744	0.00024	-2.6326	-1.486910	
4.4219	-0.9588	-1.221	-2.199	2.8253	-1.050042	
-3.2526	3.1026	2.0126	0.8726	-3.6625	0.064960	
-0.3602	-2.6395	0.0084	4.4757	0.4209	0.701331	
2.8747	-0.0487	2.7356	-0.6006	-2.0961	-0.581925	
-1.6155	2.9991	1.8965	3.1341	1.3557	-0.986742	
2.5162	-2.2655	-2.6032	-2.9577	2.4228	3.291165	
0.8845	-2.4833	1.3516	-4.0021	2.2128	0.975319	
-1.3574	2.8777	-4.0362	-2.4598	-2.8287	3.208463	

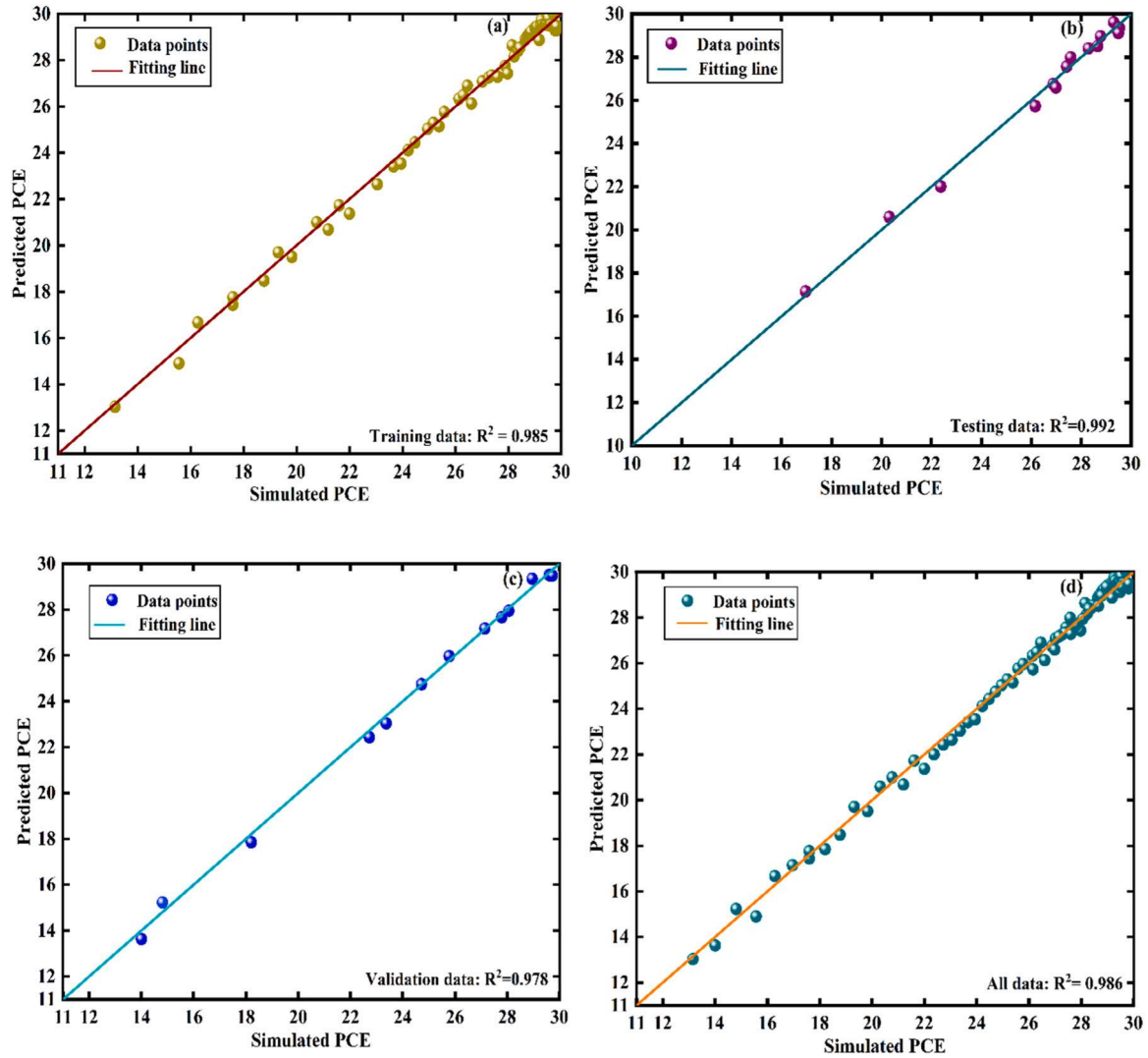


Fig. 18. Regression plot between simulated and predicted PCE for the LED spectrum: (a) training, (b) testing, (c) validation, and (d) all data.

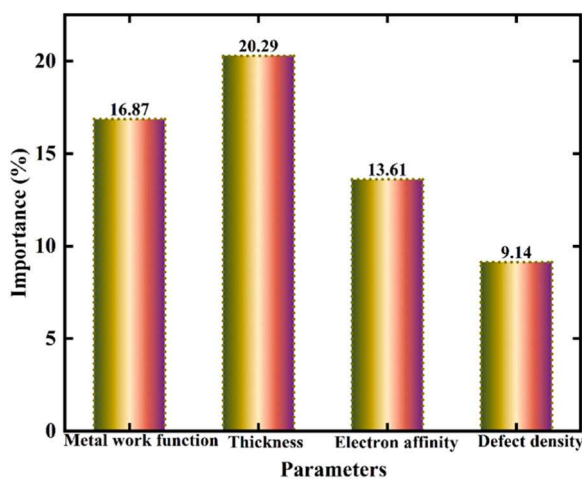


Fig. 19. Importance of parameters in the ANN model for the LED spectrum.

evaluated using these agitated inputs. The results showed only minor variation in PCE predictions, with changes well within acceptable error margins. The performance metrics R^2 , MSE, and MSE remained stable, indicating that the model is resilient to small fluctuations in input data.

These findings confirm that the ANN model maintains a high level of predictive reliability under realistic variations, making it suitable for practical applications where data noise is inevitable.

Fig. 16 (a-d) presents a graphical comparison between the SCAPS-simulated and ANN-predicted PCE under the sunlight spectrum. Specifically, Fig. 16 (a) shows the training data with predicted and simulated PCE values in the range of 6–19 %, Fig. 16 (b) presents the testing data from 11 % to 19 %, Fig. 16 (c) shows the validation data with PCE values ranging from 6 % to 20 %, and Fig. 16 (d) depicts the overall data with the values from 7 % to 19 %. The performance metrics of simulated PCE versus ANN prediction PCE are as follows: 0.964 for training, 0.973 for testing, 0.959 for validation, and 0.968 for overall, illustrating a strong correlation between the simulated and predicted PCE across all datasets. Furthermore, the PCE of the four parameters (thickness, defect density, electron affinity, and metal work function) based on HTL-free PSCs, predicted by the ANN model under the sunlight spectrum, closely aligns with the PCE values obtained using the SCAPS methodology, demonstrating excellent agreement and reliability.

Fig. 17 shows the importance of various parameters in the HTL-free PSC device. Using this ANN prediction model, absorber thickness is found to be the most important parameter under the sunlight spectrum, with a contribution of 19.38 %. The metal work function, defect density, and electron affinity contribute of 15.27 %, 13.14 %, and 12.61 %, respectively. Based on these results, absorber thickness emerges as the

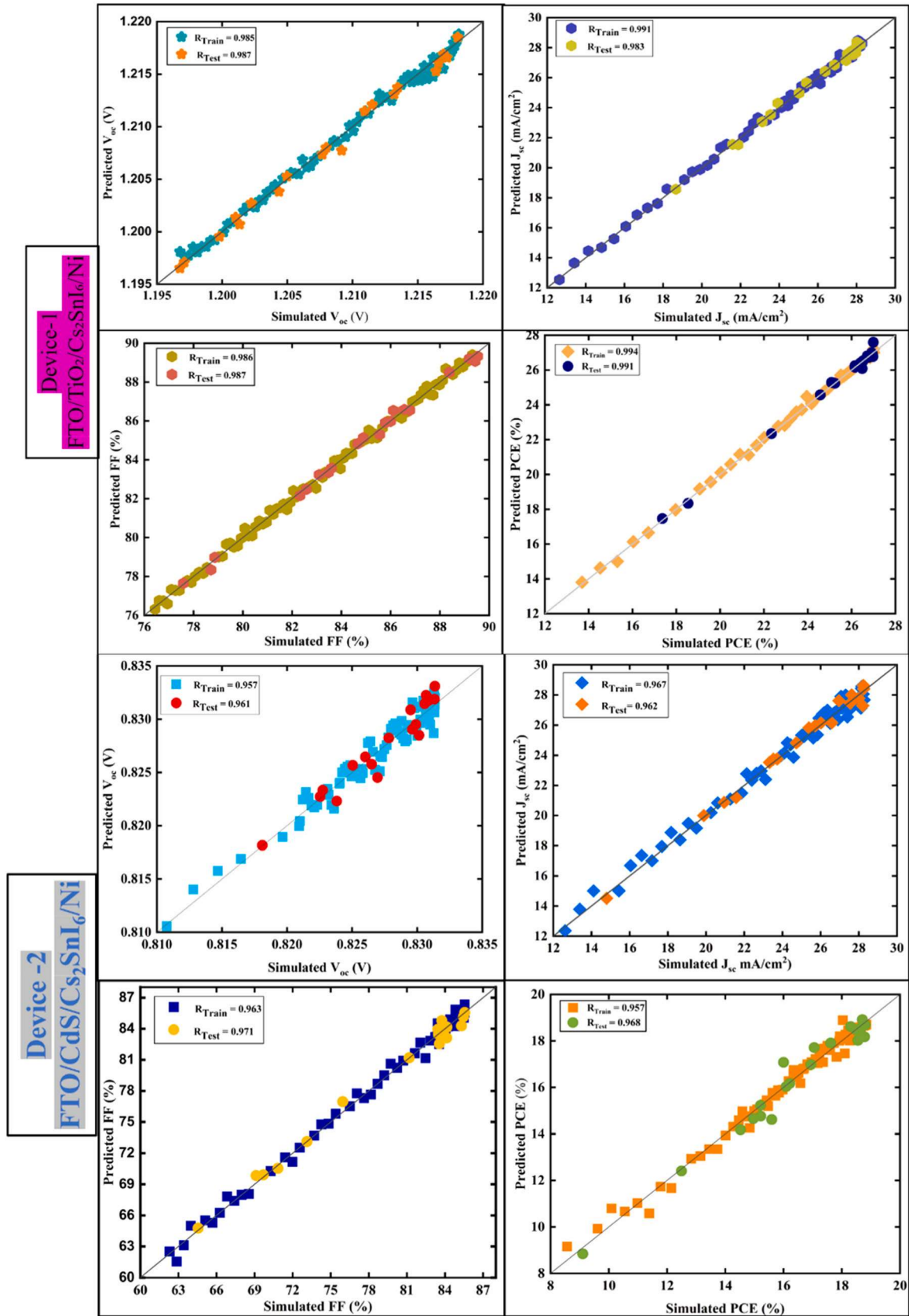


Fig. 20. Regression plot between simulated and predicted V_{oc} , J_{sc} , FF, and PCE for the sunlight spectrum with comparison between Device -1 and Device -2.

Table 9
Comparison of Optimized Devices with ANN Prediction Model and SCAPS Simulation under sunlight spectrum.

Parameters	Device-1 FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni (ANN Prediction)	Device-1 FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni (SCAPS Simulation)	Device-2 FTO/CdS/Cs ₂ SnI ₆ /Ni (ANN Prediction)	Device -2 FTO/CdS/Cs ₂ SnI ₆ /Ni (SCAPS Simulation)
V _{oc} (V)	R _{Train} = 0.985 R _{Test} = 0.987	1.11	R _{Train} = 0.957 R _{Test} = 0.961	1.03
J _{sc} (mA/cm ²)	R _{Train} = 0.991 R _{Test} = 0.983	26.86	R _{Train} = 0.967 R _{Test} = 0.962	25.86
FF (%)	R _{Train} = 0.986 R _{Test} = 0.987	82.26	R _{Train} = 0.957 R _{Test} = 0.968	76.72
PCE (%)	R _{Train} = 0.994 R _{Test} = 0.991	27.01	R _{Train} = 0.963 R _{Test} = 0.971	23.84

most significant parameter in this PSC device.

The same network configuration of the ANN is also applied to LED light spectrum data. The very low MSE and RMSE values indicate that the model has high prediction accuracy for the HTL-free PSC device under LED light conditions. Table 7 presents a comprehensive performance summary of the ANN model under the LED spectrum, emphasizing its accuracy. Table 8 presents the weights, biases, and parameters of the trained ANN model, which are essential for predicting PCE values under the LED spectrum.

Fig. 18 (a-d) illustrates the ANN predictive model applied to the indoor LED spectrum. The figure compares the predicted and simulated PCE values. Specifically, Fig. 18 (a-d) displays the training, testing, validation, and total data with values ranging from 10 % to 30 %. The prediction evaluation model results show an accuracy of 0.985 for training, 0.992 for testing, 0.978 for validation, and 0.986 overall. These results indicate a strong and close alignment between the SCAPS-simulated values and the ANN-predictive model, demonstrating the model's accuracy. Additionally, the model demonstrates excellent accuracy with an R-squared value close to 1, indicating a strong correlation between the predicted and simulated results.

Fig. 19 shows the importance of the three input parameters—absorber thickness, absorber defect density, electron affinity, and work function of back contact metal. Among these, the absorber thickness has the highest importance, with a value of 20.29 %, which is followed by the metal work function at 16.87 %, electron affinity at 13.61 % and defect density at 9.14 %. The SCAPS simulations using the LED indoor spectrum align well with the ANN prediction model, indicating very good agreement.

Fig. 20 shows the comparison between Device-1 (FTO/TiO₂/Cs₂SnI₆/Ni) and Device-2 (FTO/CdS/Cs₂SnI₆/Ni) using the ANN prediction model. Previously, we identified thickness as the most critical parameter for HTL-free devices under the solar light spectrum. Based on this key parameter as the input data, we compare these two devices to predict the highest-performing and most reliable device for the HTL-free configuration. The comparison predicts the performance metrics V_{oc}, J_{sc}, FF, and PCE for the PSC devices. For Device-1 (FTO/TiO₂/Cs₂SnI₆/Ni), the predicted V_{oc} values are 0.985 for training and 0.987 for testing. Similarly, the J_{sc} values are 0.991 for training and 0.983 for testing. Additionally, the FF shows values of 0.986 for training and 0.987 for testing. Finally, the PCE values are 0.994 for training and 0.991 for testing. The SCAPS model for Device-1 closely matches the predicted ANN model. On the other hand, Device-2 (FTO/CdS/Cs₂SnI₆/Ni) is also investigated under the solar light spectrum. For this device, the four metrics show the following values: V_{oc} is 0.957 for training and 0.961 for testing; J_{sc} is 0.967 for training and 0.962 for testing; FF is 0.957 for training and 0.968 for testing; and PCE is 0.963 for training and 0.971

for testing. Based on the ANN prediction model, Device-1 (FTO/TiO₂/Cs₂SnI₆/Ni) performs and demonstrates greater reliability under the solar light spectrum compared with Device-2 (FTO/CdS/Cs₂SnI₆/Ni). Table 9 provides a detailed comparison between Device -1 and Device-2 based on SCAPS analysis and the ANN prediction model under the solar spectrum. The prediction analysis of the four metrics (V_{oc}, J_{sc}, FF, and PCE) shows higher values for Device -1 than for Device-2. This table demonstrates that Device-1 outperforms Device -2 in both prediction and simulation.

Fig. 21 presents the ANN prediction results comparing Device-1 (FTO/TiO₂/Cs₂SnI₆/Ni) and Device-2 (FTO/CdS/Cs₂SnI₆/Ni) using the LED spectrum based on the thickness input parameter. Device-1 (FTO/TiO₂/Cs₂SnI₆/Ni) presents V_{oc} of 0.986 and 0.991 for the training and testing datasets, respectively. Similarly, the J_{sc} and FF under LED spectrum conditions show training and testing values of 0.987 and 0.993 and 0.984 and 0.990, respectively. The predicted PCEs for Device-1 are 0.993 and 0.995 for training and testing, respectively.

On the other hand, Device-2 (FTO/CdS/Cs₂SnI₆/Ni) gives V_{oc} of 0.962 and 0.967 for training and testing, respectively. The corresponding values for J_{sc} and FF are 0.951 and 0.965, and 0.973 and 0.967 for training and testing, respectively. The predicted PCEs for Device-2 are 0.960 and 0.957 for training and testing, respectively. Table 10 demonstrates the optimized device comparison using the ANN model and SCAPS simulation under the LED spectrum. Based on the hybrid model combining the ANN predictive model and SCAPS, Device 1 shows better performance.

3.9. Limitations and future work

While this study provides valuable insights into the optimization of HTL-free Cs₂SnI₆ perovskite solar cells using SCAPS-1D simulations and artificial neural network (ANN) modeling, several limitations should be acknowledged:

3.9.1. Simulation assumptions

The study is based on SCAPS-1D simulations, which rely on idealized assumptions regarding material properties and device structure. While these assumptions are reasonable for preliminary analysis, they do not account for real-world complexities such as nonuniformities in film deposition, fabrication defects, and environmental factors. Experimental validation of the simulated results is necessary to confirm the performance predictions and better understand the discrepancy between ideal simulations and actual device behavior.

3.9.2. LED spectrum data

In this study, the LED light spectrum data used in the simulations were sourced from existing literature. Although this approach provided valuable insights into the performance of the HTL-free perovskite solar cell under LED illumination, it does not account for the full range of possible variations in LED spectra. Future studies could include custom LED spectra or real-world testing to examine the impact of variations in light source characteristics on device performance.

3.9.3. Material and model selection

The materials and model configurations used in this study—such as Ni for the back contact and a 4–12–4 ANN architecture—were selected based on their initial performance in simulations. However, alternative materials (e.g., graphene oxide for ETL or gold for back contact) or more complex ANN architectures may lead to enhanced performance. Exploring a wider range of materials and model configurations could provide a more comprehensive understanding of the optimal device design.

3.9.4. Absence of real-world device testing

While this work focused on simulating device performance under various conditions, it did not incorporate real-world device fabrication

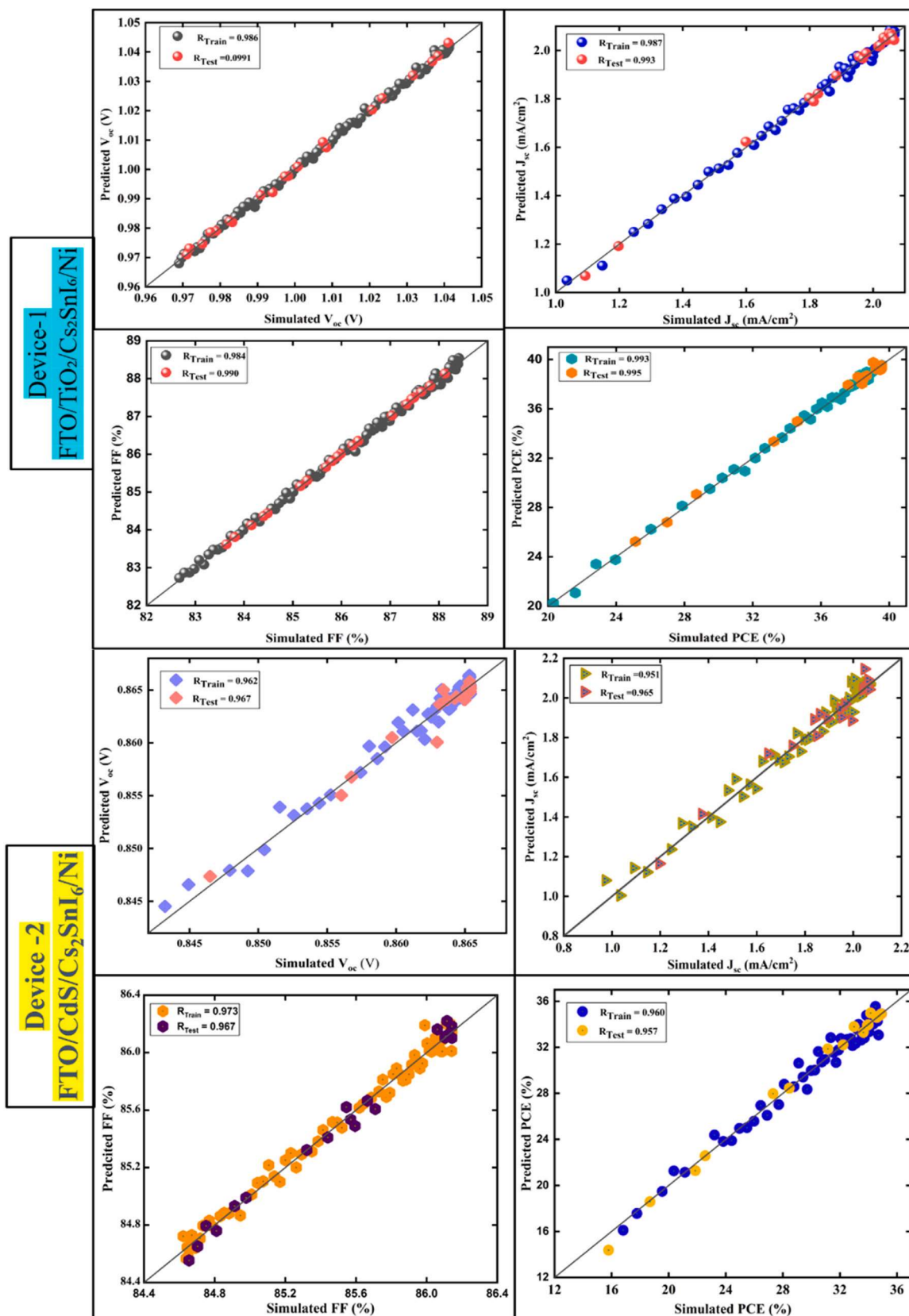


Fig. 21. Regression plot between simulated and predicted V_{oc} , J_{sc} , FF, and PCE for the LED spectrum with comparison between Device-1 and Device-2.

Table 10
Comparison of Optimized Devices with ANN Prediction Model and SCAPS Simulation under LED spectrum.

Parameters	Device-1 FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni (ANN Prediction)	Device-1 FTO/TiO ₂ /Cs ₂ SnI ₆ /Ni (SCAPS Simulation)	Device-2 FTO/CdS/Cs ₂ SnI ₆ /Ni (ANN Prediction)	Device –2 FTO/CdS/Cs ₂ SnI ₆ /Ni (SCAPS Simulation)
V _{oc} (V)	R _{Train} = 0.986 R _{Test} = 0.991	1.0097	R _{Train} = 0.962 R _{Test} = 0.967	0.866
J _{sc} (mA/cm ²)	R _{Train} = 0.987 R _{Test} = 0.993	2.01	R _{Train} = 0.951 R _{Test} = 0.965	2.06
FF (%)	R _{Train} = 0.984 R _{Test} = 0.990	83.72	R _{Train} = 0.973 R _{Test} = 0.967	86.14
PCE (%)	R _{Train} = 0.993 R _{Test} = 0.995	38.75	R _{Train} = 0.960 R _{Test} = 0.957	34.88

or testing. Scaling the HTL-free Cs₂SnI₆ perovskite solar cell to larger-area modules or evaluating long-term stability in actual operational environments were not addressed in this study. These factors are critical for the practical deployment of the technology.

3.9.5. Future work

In light of the aforementioned limitations, several directions for future research are proposed:

3.9.6. Experimental validation

Future research should focus on experimental validation of the simulation results, particularly with real-world devices. This could include fabricating HTL-free Cs₂SnI₆ perovskite solar cells and testing their performance under both solar and LED light spectra, along with long-term stability tests.

3.9.7. Alternative materials and device configurations

Further studies should explore alternative materials for the ETL and back contact. For example, mesoporous TiO₂ or graphene oxide could be considered for the ETL, while alternative metals like gold or platinum could be tested for the back contact to enhance device stability. Additionally, more advanced ANN models (e.g., deep learning or ensemble methods) could be investigated to improve prediction accuracy.

3.9.8. Scalability and upscaling

Research into the scalability of HTL-free Cs₂SnI₆ perovskite solar cells is necessary to evaluate their potential for large-area applications. This includes assessing the challenges associated with scaling up the fabrication process, ensuring uniform film deposition, and mitigating issues such as increased series resistance and nonuniformity.

3.9.9. Electron affinity engineering

While this study simulated electron affinity optimization through SCAPS-1D, future work should focus on experimentally achieving controlled electron affinity tuning in Cs₂SnI₆ layers. This will help understand the practical challenges, such as potential trap states or degradation, which are not captured in simulations.

4. Conclusion

This study investigates the performance of an HTL-free perovskite solar cell (PSC) device, focusing on the FTO/TiO₂/Cs₂SnI₆/Ni structure with validated SCAPS and ANN models. By utilizing electron affinity

engineering, the selection of ETL materials is optimized, leading to a highly efficient PSC device. The device performance is evaluated under two light conditions: the solar light spectrum (representing outdoor illumination) and the LED light spectrum (representing indoor lighting). Initially, under the solar spectrum, the device achieves an efficiency of 18.20 %, while under the LED spectrum, it reaches 28.94 %. Further optimization of key parameters absorber thickness, absorber defect density, electron affinity, and work function of back contact metal results in the following values: 600 nm thickness, 10¹³ cm⁻³, 3.9 eV electron affinity, and Ni (5.5) work function for the solar spectrum, and 1100 nm thickness, 3.7 eV electron affinity, and Ni (5.5 eV) back contact metal work function for the LED spectrum. The optimized device demonstrates a V_{oc} of 1.11 V, J_{sc} of 26.86 mA/cm², FF of 82.16 %, and PCE of 27.01 % under the solar spectrum. Under the LED spectrum, the device shows a V_{oc} of 1.00 V, J_{sc} of 2.01 mA/cm², FF of 83.72 %, and PCE of 38.75 %. The ANN-based predictive model confirms the accuracy of the PCE results, showing a strong correlation with SCAPS simulations. The predictive model is able to estimate the efficiency with the MSE of 8.127 × 10⁻⁴, the RMSE of 0.00944, and the R² value of 0.968 under the solar light spectrum. For the LED light spectrum, the MSE is 3.6585 × 10⁻⁴, the RMSE is 0.00614, and the R² is 0.986. This investigation reveals that the absorber thickness is the most critical parameter, followed by other key parameters. Finally, we compare optimized Device-1 (FTO/TiO₂/Cs₂SnI₆/Ni) and Device-2 (FTO/CdS/Cs₂SnI₆/Ni) using the ANN prediction model with PSC performance metrics. Based on the ANN prediction model under both light spectra, Device-1 performs better than Device-2. In conclusion, the HTL-free PSC device developed in this study demonstrates significant potential for high-efficiency solar cells. The HTL-free architecture simplifies fabrication and reduces cost, supporting the scalability of PSCs. In addition, ANN-based modeling accelerates optimization and minimizes experimental effort, offering a commercially relevant pathway for PSC development. The combination of SCAPS simulations with ANN-based predictions offers a robust approach to optimizing PSCs, positioning this device as a promising candidate for further development and commercialization in renewable energy technologies. To the best of our knowledge, this is the first study to integrate SCAPS-1D simulation and ANN modeling for performance optimization of HTL-free Cs₂SnI₆-based PSCs, offering a novel data-driven pathway for high-efficiency device design. These findings suggest that the optimized PSC structure holds strong potential for both indoor (e.g., Internet of Things (IoT) and low-light devices) and outdoor (e.g., rooftop solar) applications, making it suitable for diverse real-world energy harvesting scenarios.

CRediT authorship contribution statement

Guobing Zhou: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Md. Munirul Hasan:** Writing – review & editing, Validation, Software. **Md Zannatul Arif:** Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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